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26 November 1981
SUPERSEDING
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9 April 1981

MILITARY SPECIFICATION
COMPUTER SET, DIGITAL
AN/ASQ-114(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall perform the functions of the digital data processor in an integrated airborne system.

1.2 Classification - The Digital Computer Set, consisting of the items listed in 6.8, shall be classified in accordance with the following nomenclature and characteristics (see 6.6):

a. Under the following conditions, the computer set nomenclature shall be the AN/ASQ-114(V):

(1) The bootstrap memory shall be a 512 word core rope, nondestructive readout (NDRO) memory (see 3.5.1.5.1.8).
(2) The number of input channels shall be 12 (see 3.5.3.1).
(3) The memory unit shall have interconnection cables between the sense amplifier maintenance modules and the memory stack maintenance modules (see 3.2.3.1). The interconnection cables are exclusive of the wire wrapped memory unit interconnection panel (see 3.5.2.8).

b. Under the following conditions, the computer set nomenclature shall be the AN/ASQ-114(V)3:

(1) The bootstrap memory shall be a 1024 word NDRO Read Only Memory (ROM) (see 3.5.1.5.1.8).
(2) The number of input channels shall be 16 (see 3.5.3.1).
(3) The memory unit shall have interconnection cables between the sense amplifier maintenance modules and the memory stack maintenance module (see 3.2.3.1). The interconnection cables are exclusive of the wire wrapped memory unit interconnection panel (see 3.5.2.8).

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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c. The computer set nomenclature shall be AN/ASQ-114(V) 4;

(1) The bootstrap memory shall be a 1024 word ROM (see 3.5.1.5.1.8).
(2) The number of input channels shall be 16 (see 3.5.3.1).
(3) The memory unit interconnections between the sense amplifier maintenance modules and the memory stack maintenance module (see 3.2.3.1) shall be through the wire wrapped interconnection panel (see 3.5.2.8).

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400J - Electronic Equipment, Aircraft, General Specification for

MIL-T-5422E - Testing, Environmental, Aircraft Electronic

MIL-I-6181 - Interference, Control Requirements, Aircraft Equipment
6 Oct 1967

MIL-C-6781B - Control Panel; Aircraft Equipment, Rack or Console Mounted

MIL-M-7793C - Meter, Time Totalizing

MIL-E-17555F - Electronic and Electrical Equipment, and Associated Repair Parts, Preparation for Delivery of

MIL-T-18303B - Test Procedures: Preproduction, Acceptance for Aircraft Electronic Equipment, format for

MIL-N-18307C - Nomenclature and Nameplates for Aeronautical Electronics and Associated Equipment

MIL-C-81511A - Connector, Electric, Circular, High Density, Quick Disconnect, Environment Resisting, General Specification for

STANDARDS

Federal

FED-STD-595A - Colors

Military

MIL-STD-275B - Printed Wiring for Electronic Equipment
Change 1 -
24 June 1966

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MIL-STD-415B - Test Points and Test Facilities Design Standard for

MIL-STD-470 - Maintainability Program Requirements (for Systems
21 Mar 1966 and Equipments)

MIL-STD-471 - Maintainability Demonstrations
15 Feb 1966

MIL-STD-480 - Configuration Control, Engineering Changes, Deviations
30 Oct 1968 and Waivers

MIL-STD-681B - Identification Coding and Application of Hook up and
Lead Wire

MIL-STD-704A - Electric Power, Aircraft, Characteristics and Utilization of
Change 1 -
7 Feb 1968

MIL-STD-756A - Reliability Prediction

MIL-STD-781B - Test Levels and Accept/Reject Criteria for Reliability
of Non-Expandable Electronic Equipment

MIL-STD-785 - Requirements for Reliability Program (for systems and
30 June 1965 equipments)

MIL-STD-794A - Parts and Equipment, Procedures for Packaging and
Packing of

MIL-STD-1130 - Connections, Electrical, Solderless Wrap
12 Nov 1965

HANDBOOK

Military

MIL-HDBK-217A - Reliability Stress and Failure Rate Data for Electronic
Equipment

PUBLICATIONS

Naval Air Systems Command

AR-5 - Microelectronics Devices Used in Avionic Equipment,
1 Oct 1966 Procedures for Selection and Approval of

Naval Sea Systems Command

WR-101 - Weapon Requirement for Advanced ASW Electronic Systems
3 Mar 1966
Amend 1 - Part 1 - Electromagnetic Control Requirements for Advanced
15 Feb 1968 ASW Avionics Systems

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Naval Sea Systems Command (continued)

WS-6118 - Design Specification for Wire, Electrical Wire Wrap Insulated and Uninsulated

WS-6119 - Unisolated Process Specification for Connection, Electrical Solderless Wrapped, Special Projects Office Naval Department

(Copies of Specifications, Standards, Drawings, and Publications required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In doing so, the following shall govern:

a. Microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.4).

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including tubes, transistors and diodes) other than microelectronic devices, shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

*3.2.3 Maintenance Module - The Maintenance Module is defined as the least complex segment of the Digital Computer Set which is capable of being removed and replaced at orgainizational level maintenance using tools, support equipment and procedures approved for the end article(s). Further, it is the segment (three maintenance modules or less) to which the contractor will provide fault detection and isolation techniques consistent with scheduled and unscheduled organizational level maintenance requirements.

3.2.3.1 Maintenance Module Requirements - There shall be no more than seven classes of Maintenance Modules as follows:

a. Type 1 Maintenance Module shall contain only microelectonic integrated packages.

b. The Type 2 Maintenance Module shall have the same dimensions and form factor as the Type 1 Maintenance Module and contains discrete components, thin film hybrid circuits, or a mixture of discrete components, thin film and microelectronic integrated packages.

c. The Type 3 Maintenance Module shall be used exclusively in the memory section of the computer system. This module shall contain no more than 4096 words of core memory, 30 bits per word.

d. The Type 4 Maintenance Module shall consist of a modular, pluggable segment of the power supply. The active power supply circuit elements with a higher probability of failure shall be packaged in the Type 4 modules.

e. The Type 5 Maintenance Module shall consist of a hard-wired memory unit containing up to 512 words of NDRO core rope memory.

f. The Type 6 Maintenance Module shall consist of a read only memory (ROM) unit containing up to 1024 words of integrated circuit memory.

g. The Type 7 Maintenance Module shall consist of an interconnect module containing no electrical components.

3.2.3.2 Computer Unit Assembly (CUA) - A Computer Unit Assembly shall be defined as a group of Maintenance Modules interconnected to form a physical unit. A CUA shall not weigh more than 40 pounds.

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3.2.3.2.1 Computer Unit Assembly Requirements - Each CUA shall have the following features:

a. It shall be capable of being repaired in the aircraft by replacement of plug-in Maintenance Modules. The removal of a Maintenance Module (except power supply Type 4 module) shall not require the removal of any adjacent modules.

b. Its weight shall not exceed 40 pounds.

*3.2.3.2.2 Special Computer Unit Assembly Requirements - The Maintenance Console CUA shall be considered a Special Unit Assembly and shall contain hard-wired switches and indicators and pluggable maintenance modules (Type 1 or Type 2).

*3.2.3.3 Digital Computer Set - The Digital Computer Set also referred to as the equipment shall consist of an Isolation Power Transformer, a Mounting Rack and the following Computer Unit Assemblies (CUA) - Central Processing Unit, Memory Units, Power Supply Unit, Input/Output Unit. The Maintenance Console shall be a separate unit and can be operated remote with a cable not more than 15 feet in length.

*3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 and MIL-STD-275 for design, construction and workmanship, except as otherwise specified herein. Exceptions to MIL-STD-275 are as follows:

a. Clinched Leads - Clinched leads shall display a visible outline of the lead on the circuit side terminal area. The lead shall not extend off the terminal area by more than one terminal area diameter. The lead shall not extend to within 0.010 inch of any adjacent circuitry and shall not protrude more than 0.015 inch from the circuit side board surface. On insulated leads, solder must wick to the top of the plated through-hole but need not exhibit good wetting on the component side.

b. Unclinched Leads - Acceptable depth of insertion shall be indicated by the visible outline of the lead on the circuit side of the board. Where the lead outline is obscured by solder on the circuit side of the board, the insertion depth shall be considered adequate if the component lead was properly prepared.

c. Lead - A lead shall be trimmed to provide a protrusion from flush to 0.015 inch before soldering and component and lead are not raised.

d. Raised Component - A raised component is one that is not mounted visually flush to the spacer, tape, adhesive, or board as applicable or is not visually parallel to the board surface.

e. Raised Lead - A raised lead is one whose axis is greater than parallel plus 5 degrees to the surface of the board.

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*3.3.1 Total Form Factor - The total weight of the equipment with 65,536 words of memory including the mounting rack (also including the maintenance console cables, handles and front air filters) shall not exceed 395 pounds. The volume of the equipment shall not exceed nine (9) cubic feet. Permanent handles shall protrude a maximum of 2.25 inches beyond the face-plate of all computer unit assemblies except the maintenance console where the handle shall be mounted on top of the unit, and this handle shall not extend more than 2.25 inches above the top of the case. Removable handles on the chassis shall extend a maximum of 2.75 inches beyond the face-plates of the repsective chassis.

*3.3.1.1 Color - The color of the equipment shall conform to FED-STD-595 color number 36555.

*3.3.1.2 Automatic Wire Wrap - The equipment may use automatic wire wrap techniques in accordance with MIL-STD-1130 or WS-6118 and WS-6119. Where used, the requirement of color coding signal wires in accordance with MIL-STD-681 is not applicable.

3.3.2 Reliability - A reliability assurance program in accordance with MIL-STD-785 shall be established and conducted.

*3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 5,000 operating hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal operation.

*3.3.2.2 Operating Life - The equipment shall have a total operating life of 50,000 hours or ten years, whichever occurs first, with reasonable servicing and replacement of parts. Operating life is defined as that period of time during which the equipmemt has an MTBF equal to, or greater than the minimum acceptable MTBF. A list of parts requiring scheduled replacement and the replacement interval shall be made available by the contractor.

3.3.2.3 Specified Mean Time Between Failure - The specified Mean-Time Between Failure (MTBF) of the digital computer set shall be 111 hours as determined by reject criteria for Test Plan XXIX of MIL-STD-781 dated 15 November 1967, when tested as specified in 4.4.3.2.

*3.3.2.3.1 Reliability Analysis - An analytical reliability model shall be prepared in accordance with the procedures of MIL-STD-756 and MIL-HDBK-217. Supplier component parts data may be used for component parts data not included in MIL-HDBK-217 or in cases where the data in MIL-HDBK-217 is not considered current. Written justification for these deviations shall be made available to the procuring activity. For production, an update of the reliability analysis model shall also be made available.

3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with MIL-M-7793. A time totalizing meter shall be included in the Power Supply Unit, the Central Processor Unit, the I/O Unit, each Memory Unit, and the Maintenance Console.

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3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with WR-101, Part I, Amendment I, and MIL-C-81511.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of WR-101, Part I, Amendment 1. The external wiring may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. Reference designations and types of connectors shall be as specified in Table I. Interconnecting cables and those portions of the connectors attached to the cables shall not be supplied as part of the equipment.

TABLE I. Connectors.

Reference Designations	Connector Type	Function
A1-J65 through A1-J80	M81511/01EF01P1	Data Input
A1-J81 through A1-J84	M81511/01EF01P2	Data Output
A1-J85 through A1-J88	M81511/01EF01P3	Input/Output Control
A8J01	M81511/01EF01S2	Maint. Console Input
A8J02	M81511/01EF01S3	Maint. Console Input
A8J03	M81511/01EF01S4	Maint. Console Input
A8J04	M81511/01EF01S5	Maint. Console Input
A8J05	MS3102R14S-6P	Real-Time Clock Input
A8J06	MS3102R20-15P	A.C. Power Input
A8J07	CA3102R24-2008-24S	A.C. Power Out
A9J01	M81511/01EF01P4	Maint. Console Out
A9J02, A9-J03, A9-J04	M81511/01EF01P1	Maint. Console Out
A9-J05	M81511/01EF01P5	Remote Control In
A9-J06	MS3120E14-5P	D. C. Power In
A10-P01	CA06EA24A24P	Power Isolation Transformer Input

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781, except that the panel shall be grey as specified in FES-STD-595, (Color No. 36231). The configuration of all control panels must be made available for approval by the procuring activity prior to preproduction testing and data delivery.

3.3.5 Interchangeability of Reordered Equipment - For reordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

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3.3.6 Interference Control - The equipment shall conform with all of the requirements of WR-101, Part I, Amendment I for design, construction, and workmanship. A control plan will be prepared in accordance with WR-101, Part I, Amendment I.

3.3.7 Maintainability - For computer module definition, see 3.2.3.

3.3.7.1 Maintainability Program - The contractor shall establish and conduct a maintainability program in accordance with MIL-STD-470 and MIL-STD-471. All maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

(a)	Civilian education:	High School graduate or equivalent.
(b)	Technical training:	Navy, Class A Technician School appropriate to the maintenance task.
(c)	Experience:	Two (2) years of technical experience in addition to technical training.

*3.3.7.1.1 Maintenance Philosophy - The maintenance philosophy for the Digital Computer Set shall be that of fault isolation of the Maintenance Module (MM) and repair by MM replacement at the organizational maintenance level. Fault isolation shall be computer initiated utilizing a diagnostic program and documentation. Use of general purpose test equipment to support self-test equipment is permitted. It shall be possible in 95 percent of the failures to fault isolate, repair by module (MM) replacement and checkout the equipment within 30 minutes without the use of special support equipment. In addition, the capability for performance testing shall be built into each system, subsystem, and equipment to provide preflight readiness indicators or when an out-of-tolerance condition is suspected.

3.3.7.1.1.1 Packaging and Modularity Requirements - The equipment shall be packaged in modular form so that, to the maximum extent practical, all subassemblies shall be the plug-in type and should be easily removable. If a special tool is required for removal, it shall be provided with each equipment.

3.3.7.1.1.2 Module Commonality - The number of different modules shall be kept to an absolute minimum. This standardization is emphasized to ease the testing and spares requirements at all levels of maintenance (see 3.2.3).

*3.3.7.1.1.3 Module Accessibility - Module arrangement shall be such that access to each module (except power supply module) does not require the removal of adjacent modules or parts other than access panels.

3.3.7.1.1.4 Keying - Modules and connectors shall be keyed to prevent the insertion of a module or connector into an improper location within or on the equipment.

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3.3.7.1.1.5 Nonrepairable Assemblies - All design for nonrepairable assemblies shall be made available to the Procuring Activity, for approval, prior to release for fabrication.

*3.3.7.1.1.6 Adjustments - Procuring Activity approval is required for design and use of adjustment, alignment or calibration functions except for those specified herein. Adjustments shall be capable of being made in the aircraft using simple hand-tools or General and Standard test equipment. Adjustments shall be identified and accessible without a requirement to remove or relocate the module. This paragraph does not apply to factory adjustments.

3.3.7.1.2 Fault Isolation -

3.3.7.1.2.1 Fault Isolation - In conjunction with, but not necessarily as part of the Performance/Readiness testing features, there shall be features incorporated which will permit fault isolation to the maintenance module (see 3.2.3) level. These features shall include the necessary switching arrangement with the Maintenance Console that will clearly indicate whether or not the function selected is within minimum performance standards. If a built-in readout device is not feasible, means shall be provided to use general or standard test equipment, i.e., Oscilloscope, VOM, etc., carried as part of normal A/C equipment.

3.3.7.1.2.2 Diagnostic Ability - The Digital Computer Set shall be logically and electrically designed to facilitate the location of malfunctioning maintenance modules through diagnostic programming or other methods in conjunction with the computer consoles and other aircraft installed equipment. Test points shall be made available at the front of each unit to assist the diagnosis of malfunctions.

3.3.7.1.2.3 Diagnostic and Maintenance Routines - The contractor shall assume responsiblity for preparation of the diagnostic routines for the computer, program listing, description for the use of each program as well as preparation of the associated computer diagnostic manual.

3.3.7.1.2.4 Manual Preparation - The computer diagnostic manual to be used with the diagnostic programs in the performance of computer self testing shall be written to a comprehension ability level as described in 3.3.7.1.

*3.3.7.1.3 Test Point Requirement - All test points shall accommodate general purpose test equipment, i.e., multimeter or oscilloscope probes.

3.3.7.1.3.1 Repair Test Points - Test points shall be provided on each (CUA) and for use to determine the performance of the CUA and in the event of failure, to determine which segment within the CUA is at fault.

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3.3.7.1.3.2 Power Supply Test Points - Accessible external individual test receptacles shall be provided on the equipment to measure the output voltages. The receptacles shall be identified as the voltage normally present at the test point.

3.3.7.1.3.3 Protective Devices - Circuit breakers shall be used in lieu of fuses whenever possible. Fuses, if used, shall be of types and ratings readily available from Navy supply. Fuse holders and circuit breakers reset mechanisms shall be easily accessible from the exterior of the equipment, plainly marked as to rating and size, and shall be of the fault indicating type and have access for quick voltage checks, or be of the fault indicating type or have access for quick voltage checks.

3.3.7.2 Maintainability Assurance - Testing to demonstrate the achievement of the maintainability requirement specified herein shall be in accordance with 4.5.

3.3.7.3 Special Support Equipment - There shall be no special support equipment for support of the Digital Computer Set except as expressly authorized by the procuring activity.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

*3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 $\pm$ 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input Power Voltage	115 $\pm$ 1.0 VAC, Line-to-neutral, 400 Hz $\pm$ 1 percent, 3Ø, four wire and 27.5 $\pm$ 0.5 VDC.

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1 AX equipment, except as modified herein.

*3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements as follows:

The vibration shall consist of two scans from 5 to 500 to 5 cycles, in a time period of 15 minutes each scan. The scans shall be conducted with the following amplitudes and acceleration levels.

a. From 5 to 10 cycles - constant double amplitude of 0.08 inch.

b. From 10 to 15 cycles - constant $\pm$ 0.41g input.

c. From 15 to 50 cycles - constant double amplitude of 0.036 inch.

d. From 50 to 500 cycles - constant $\pm$ 3g input.

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In addition, the equipment shall operate satisfactorily when subjected to a constant input of 2g at 17, 68, and 136 cycles for a period of 15 minutes at each frequency.

3.3.10.2 Humidity Range - The equipment shall operate satisfactorily over the relative humidity range of 0 to 90 percent. No damage to the equipment shall occur from prolonged exposure to the atmosphere at relative humidities over the range of 0 to 100 percent.

3.3.10.3 Attitude - The equipment shall operate satisfactorily in all attitudes and orientations within $\pm$ 90° of its normal mounting orientation.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 0.5 minute under standard conditions, five minutes at -45°C, and fifteen minutes at -55°C.

*3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as defined herein. In case of a power failure, or return of power to normal, the computer shall be protected against loss of stored information from the memory. The equipment shall operate within the bounds of limits 2 and 3 of Figure 3 and within limits 5 and 6 of Figure 5 of MIL-STD-704, except limit 3 of Figure 3 shall not go below 80 volts. No damage shall be suffered by the equipment and normal operation shall automatically resume upon return of the input voltage to levels within limits 2 and 3 of Figure 3 except if the input voltage exceeds limit 1 of Figure 3. Protective steps shall be taken as required in 3.5.4.1.4. The power required shall be as indicated in Table II.

TABLE II. Power Requirements.

(1) A/C Power, Three Phase, 115/200V, Category "B".
(2) D.C. Power, 28V, Category "B".
(3) Power not to exceed the following:

Memory Size	Input Power Peak	Input Power Normal
32,768 words	1134 watts	919 watts
49,152 words	1339 watts	1064 watts
65,536 words	1461 watts	1167 watts

3.4 Performance - Unless otherwise specified herein, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail Requirements - All computer units except the Maintenance Console and the Isolation Power Transformer Assembly shall be affixed to the Mounting Rack.

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*3.5.1 Central Processor Unit - The Central Processor Unit shall meet the following requirements.

3.5.1.1 Function - The Central Processor (CP) shall contain all the arithmetic and control circuits required to perform each instruction in the instruction repertoire, and all timing within the CP shall be consistent with a two microsecond memory cycle time. The CP shall provide for a maximum memory configuration of up to 131,072 thirty (30) bit words. Overlap instruction techniques shall be used in the computer.

*3.5.1.2 Form Factor - The Central Processor shall be enclosed in a case 5.5 inches wide by 12.6 inches high by 14.4 inches long (alignment pins and handles excluded).

*3.5.1.3 Weight - The weight of the Central Processor shall not exceed 36 pounds.

3.5.1.4 Memory Addressing - The Central Processor shall contain the electronics for addressing 131,072 words of memory.

*3.5.1.5 Logic and Control - The Central Processor (CP) shall contain the circuitry to necessary execute all computer operations and to control all arithmetic and logic operations as required by the instructions repertoire.

*3.5.1.5.1 Logic Characteristics -

3.5.1.5.1.1 Address Logic - Single address logic shall be used.

3.5.1.5.1.2 Word Length - The computer instruction word shall be 30 binary digits, and the operand word shall be 30 binary digits or 15 binary digit half-words. Operations on half-word operands shall not increase instruction execution time over that for full-word operands.

*3.5.1.5.1.3 Cycle Time - The Central Processor cycle time for logic operations and instruction execution shall be multiples of two (2) microseconds $\pm$ 6 percent.

3.5.1.5.1.3.1 Master Clock Rate - The computer will have a master clock with a frequency of two megacycles $\pm$ 6 percent.

3.5.1.5.1.4 Index Registers - There shall be seven (7) index registers contained within the Central Processor Unit.

3.5.1.5.1.5 Internal Real Time Clock - The computer shall have an internal clock which shall advance the count in the real time clock memory register at a rate of 1024 counts per second. The clock accuracy shall be $\pm$ 2 counts in a period of 10 seconds.

3.5.1.5.1.6 External Real Time Clock - The computer shall provide an adaptor to connect an external real time clock to the computer. When an external clock is connected, the internal real time clock will be disconnected and the real time clock memory register shall be incremented at a rate established by the external clock. The external clock rate shall be 100 KHz per second maximum.

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*3.5.1.5.1.7 Internal Count Down Clock - The computer shall have an internal clock register which after being loaded with a positive number shall be decremented at the rate of the real time clock, internal or external, depending upon which is connected. When the register has been counted to zero, a program interrupt shall be generated.

3.5.1.5.1.8 Bootstrap Memory Unit - The Central Processor shall contain a 30-bit core rope Nondestructive Readout (NDRO) memory with up to 512 words or a 30-bit Read Only Memory (ROM) with up to 1024 words (see 6.6).

3.5.1.5.1.9 I/O Interface - The Central Processor shall interface with the I/O Unit to control up to 16 input and 16 output channels.

*3.5.1.5.2 Overlap Operation - The Central Processor shall contain the logic to address any two (2) memory units, of the possible eight (8) memory units, simultaneously to permit overlap operation. Instruction execution time for overlap instruction execution shall be as indicated in Table III. If instruction execution will result in a demand to reference a single 16K memory unit, instruction execution shall revert to nonoverlapped operation.

3.5.1.5.3 Expanded Memory Operation - The Central Processor shall provide the logic necessary to operate through the 131,072 word memory; however, programs used on the CP-823/U computer, shall be capable of being run on this computer, when the maximum memory utilized by the program is 32,768 words.

*3.5.1.5.4 Instruction Repertoire Characteristics - The computer shall be capable of executing the instruction specified in Table III, and the instruction interpretation shall be as set forth in Appendix A.

3.5.1.5.4.1 Format I Instruction Word - The Format I instruction words are used to implement non-input/output instructions and have the format shown in Figure 1. Appendix A contains information on the instructions.

3.5.1.5.4.2 Format I I/O Instruction Word - Format I input/output instruction words are used to implement input/output instructions and have the format shown in Figure 2. Appendix A contains information on the instructions.

3.5.1.5.4.3 Format II General Instruction Word - The Format II general instruction words provide additional instructions in the computer by providing instruction variations with a basic function code (f) of 77. Instruction interpretation is performed on the next lower six bits. Appendix A contains information on the instructions and the format is shown in Figure 3.

3.5.1.5.4.4 Format II Special Instruction Word - The Format II special instruction words are used to provide additional instructions in the computer by providing instruction variations with a basic function code (f) of 77, and has the format shown in Figure 4. See Appendix A for specific instruction information.

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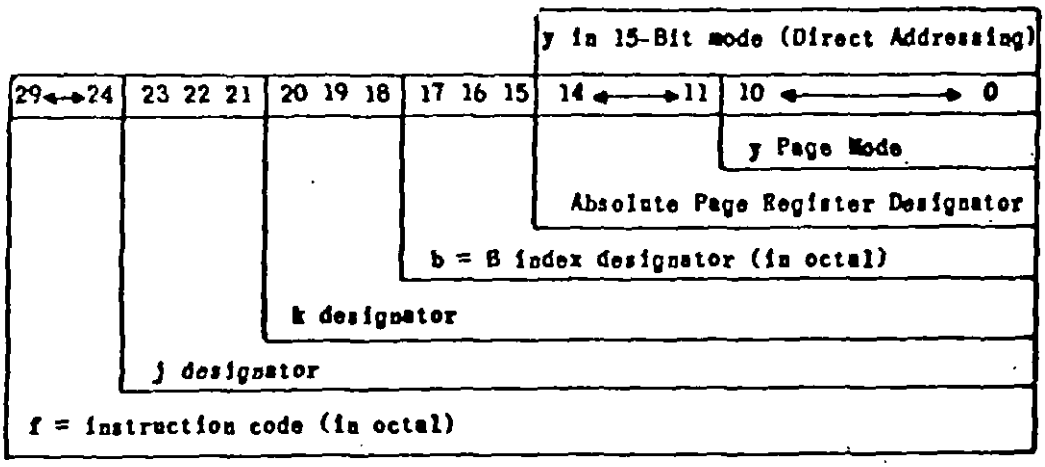


Figure 1. Format I Instruction Words

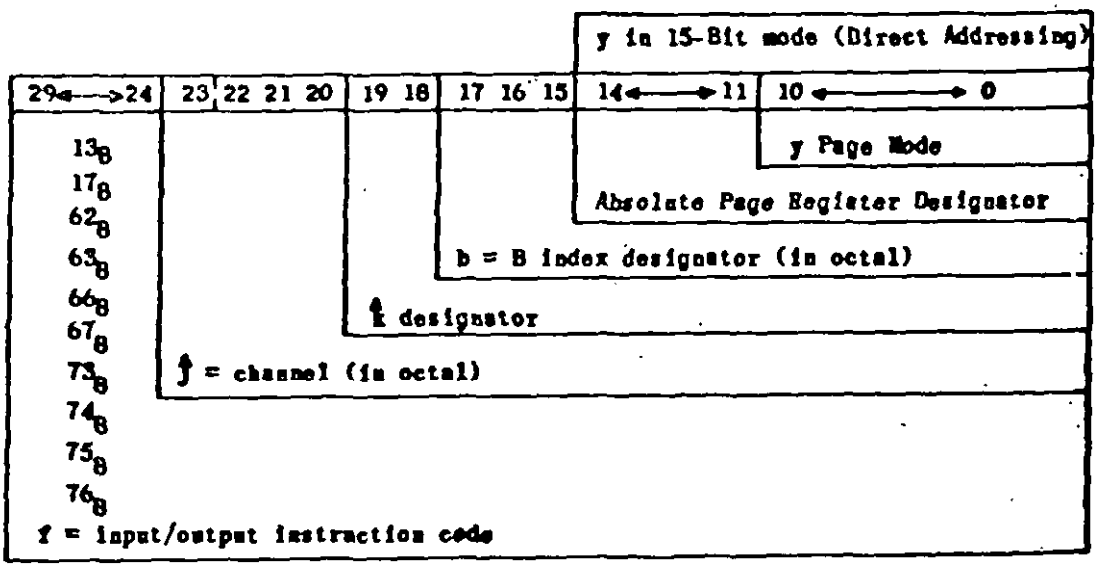


Figure 2. Format I I/O Instruction Words

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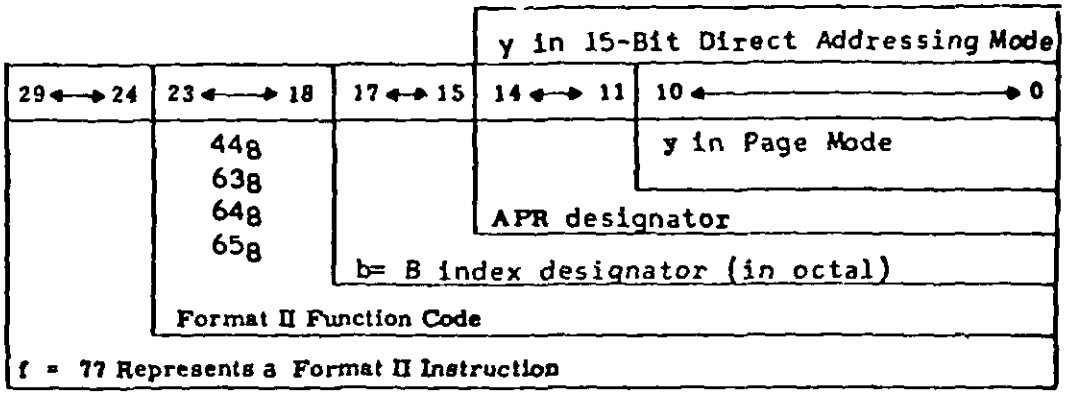


Figure 3. Format II General Instruction

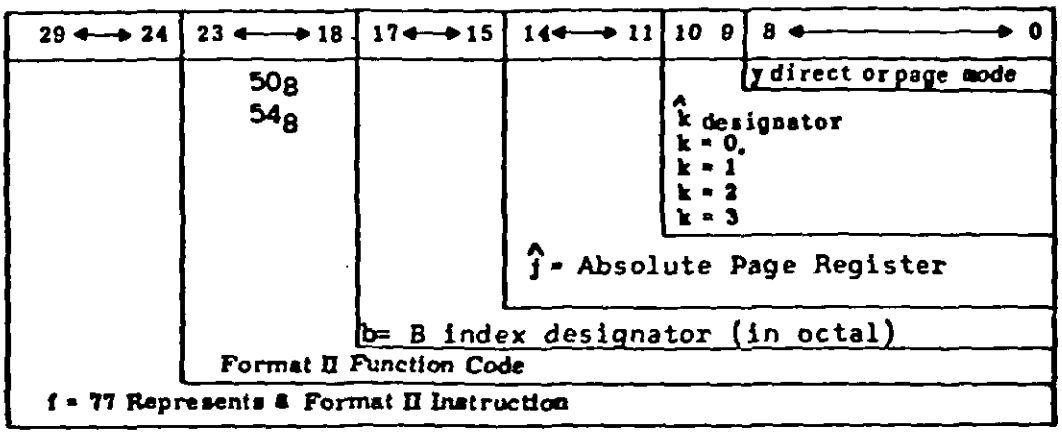


Figure 4. Format II Special Instruction

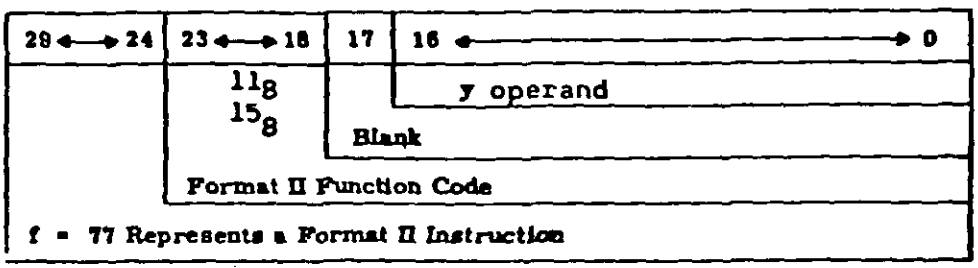


Figure 5. Format II Direct Addressing Instruction

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TABLE III Instruction Repertoire

CODE (OCTAL)	INSTRUCTION	EXECUTION TIME(S) usec					
		SAME CORE BANK j =0, 1	ALT. CORE BANK j=1	SAME CORE BANK j ≠ 0, 1		ALT. CORE BANK j ≠ 1	
				SKIP	NO SKIP	SKIP	NO SKIP
				Jump	No Jump	Jump	No Jump
.00*	Fault Interrupt	4	4	4	4	4	4
01	Right Shift Q	4-6	4-6	6-8	4-6	4-8	2-6
02	Right Shift A	4-6	4-6	6-8	4-6	4-8	2-6
03	Right Shift AQ	4-6	4-6	6-8	4-6	4-8	2-6
04	Compare	4	4	6	4	4	2
05	Left Shift Q	4-6	4-6	6-8	4-6	4-8	2-6
06	Left Shift A	4-6	4-6	6-8	4-6	4-8	2-6
07	Left Shift AQ	4-6	4-6	6-8	4-6	4-8	2-6
10	Enter Q	4	4	6	4	4	2
11	Enter A	4	4	6	4	4	2
12	Enter B	4	4	6	4	4	2
13*	External Function on C j	6 min	6 min	6 min	6 min	6 min	6 min
14	Store Q	4	4	6	4	4	2
15	Store A_n	4	4	6	4	4	2
16	Store B^n	4	4	6	4	4	2
17*	Store C j	6 min	6 min	6 min	6 min	6 min	6 min
	Test EFB	4	4	4	4	4	4
20	Add A	4	4	6	4	4	2
21	Subtract A	4	4	6	4	4	2
22	Multiply	18	18	20	18	20	18
23	Divide k ≠ 7	32	32	34	32	34	32
23	Square Root k = 7	32	32	34	32	34	32
24	Replace A + Y	6	6	6	6	4	4
25	Replace A - Y	6	6	6	6	4	4
26	Add Q	4	4	6	4	4	2
27	Subtract Q	4	4	6	4	4	2

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TABLE III Instruction Repertoire (Continued)

CODE (OCTAL)	INSTRUCTION	EXECUTION TIME(S) u sec					
		SAME CORE BANK i = 0,1	ALT. CORE BANK i=1	SAME CORE BANK i ≠ 0,1		ALT. CORE BANK i ≠ 1	
				SKIP	NO SKIP	SKIP	NO SKIP
30	Enter Y + Q	4	4	6	4	4	2
31	Enter Y - Q	4	4	6	4	4	2
32	Store A + Q	4	4	6	4	4	2
33	Store A - Q	4	4	6	4	4	2
34	Replace Y + Q	6	6	6	6	6	4
35	Replace Y - Q	6	6	6	6	6	4
36	Replace Y + Q	6	6	6	6	6	4
37	Replace Y - 1	6	6	6	6	6	4
40	Enter Logic Product	4	4	6	4	4	2
41	Add Logical Product	4	4	6	4	4	2
42	Subtract Logical Product	4	4	6	4	4	2
43	Compare Mask	4	4	6	4	4	2
44	Replace Logical Product	6	6	6	6	6	4
45	Replace A+ Logical Product	6	4	6	6	6	4
46	Replace A - Logical Product	6	4	6	6	6	4
47	Store Logical Product	4	4	6	4	4	2
50	Selective Set	4	4	6	4	4	2
51	Selective Complement	4	4	6	4	4	2
52	Selective Clear	4	4	6	4	4	2
53	Selective Substitute	4	4	6	4	4	2
54	Replace Selective Set	6	4	6	6	6	4
55	Replace Selective Complement	6	4	6	6	6	4
56	Replace Selective Clear	6	4	6	6	6	4
57	Replace Selective Substitute	6	4	6	6	6	4

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TABLE III Instruction Repertoire (Continued)

CODE (OCTAL)	INSTRUCTION	EXECUTION TIME(S) u sec					
		SAME CORE BANK i ≠ 0,1	ALT. CORE BANK i=1	SAME CORE BANK i ≠ 0,1		ALT. CORE BANK i=1	
				SKIP JUMP	NO SKIP NO JUMP	SKIP JUMP	NO SKIP NO JUMP
60	Jump (Arithmetic)			4	4	4	2
61	Jump (Manual)			4	4	4	2
62	Jump on C^n Active Input Buffer			4	4	4	4
63	Jump on C^n Active Output Buffer			4	4	4	4
64	Return Jump (Arithmetic)			6	4	6	4
65	Return Jump (Manual)			6	4	6	4
66*	Terminate C^n Input Buffer	2	2				
67*	Terminate C^n Output Buffer	2	2				
70*	Repeat	4	2				
71*	B Skip on B j			6	4	6	4
72*	B Jump on B j			4	4	4	4
73*	Input C_j	6 min	6 min	6 min	6 min	6 min	6 min
74*	Output C_j	6 min	6 min	6 min	6 min	6 min	6 min
75*	Input C_j w/monitor	6 min	6 min	6 min	6 min	6 min	6 min
76*	Output C_j w/monitor	6 min	6 min	6 min	6 min	6 min	6 min
77 11*	Enter A with Content of Address in U(0-16)	4	2				
77 15*	Store A at Address in U(0-16)	4	2				
77 44*	Test and Set Flag	6	4				
77 50*	Enter Absolute Page Register	4	4				
77 54*	Store Absolute Page Register	4	4				
77 63*	Load B^7 and Jump	4	4				
77 64*	Load B^7 and Jump-Set Direct Mode	4	4				
77 65*	Load B^7 and Jump-Set Page Mode	4	4				
77 00*	Fault	4	4				
77 77*	Fault	4	4				

* Instruction time does not vary due to the j designator.

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3.5.1.5.4.5 Format II Direct Addressing Instruction Word - The Format II direct addressing instructions provide additional instructions in the computer to direct address up to 131,072 words of memory. A variation of the function code 77 is used and the format is shown in Figure 5.

3.5.1.5.5 Program Interrupts - Provisions shall be made for the interruption of a running program by an event which may occur asynchronously with that program. An interrupt shall suspend the normal program sequence and cause the execution of the instruction located in a permanently assigned interrupt entrance address. The interrupts and their priority of execution shall be as specified in paragraphs 3.5.1.5.5.1 through 3.5.1.5.5.6.

3.5.1.5.5.1 Power On (Start) - This signal, while not an interrupt, shall be initiated with the highest priority and takes precedence over all interrupts.

3.5.1.5.5.2 Power Failure Interrupt - This interrupt shall be initiated when an out-of-tolerance input voltage transient is detected. The power failure interrupt shall be the highest priority interrupt within the computer, and it shall interrupt all equal or lower priority interrupts and lock out the initiation of all lower priority interrupts until released.

3.5.1.5.5.3 Program Fault Interrupt - This interrupt shall be initiated whenever an instruction with an illegal function code (f=00, 7700 or 7777) is executed. The program fault interrupt shall be the second highest priority interrupt; it shall interrupt all equal or lower priority interrupts and lock out the initiation of all lower priority interrupts until released.

3.5.1.5.5.4 Memory Protect Interrupt - This interrupt shall be initiated whenever an instruction is executed that initiated a memory write cycle into a protected memory area. The memory protect interrupt shall be the third highest priority interrupt; it shall interrupt all equal or lower priority interrupts and lock out the initiation of all lower priority interrupts.

3.5.1.5.5.5 Count Down Clock Interrupt - This interrupt shall be initiated when the clock memory register (000161) is decremented to zero. The count down clock shall be enabled, set to the desired time, or disabled, set to zeros, under program control. The Count Down Clock Interrupt shall be the fourth highest priority interrupt; it shall interrupt all equal or lower priority interrupts and lock out the initiation of all lower priority interrupts.

*3.5.1.5.5.6 I/O Interrupts - The I/O Interrupts shall be subdivided as follows:

a. A monitor interrupt shall be initiated when an input, output, or external function buffer with monitor terminates at the end of the transfer and an external interrupt word has been stored at the preassigned memory location for the channel. The computer will treat an external interrupt as a monitor interrupt however, the computer will use the proper monitor interrupt preassigned memory location for the channel.

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b. An intercomputer time-out interrupt shall be initiated whenever the transmitting computer has not received acknowledgement of the transmitted command or data word between computers within a specified time. The length of the time period used with the time-out may vary; therefore, a user may specify, when the order is placed, any time period value in accordance with the formula t (time-out) = $1/A \times 2^n$ second, where n may be any integer from 0 to 29, and A is the internal or external real time clock frequency.

c. An external interrupt shall be initiated when the computer has set the interrupt enable, the external device places the interrupt word on the data lines and sets the external interrupt request signal line.

I/O interrupts shall take the fifth highest interrupt priority. I/O priority shall be as defined in paragraph 3.5.3.1.2.10. Any I/O interrupt shall lock-out the initiation of all other I/O interrupts until released.

*3.5.1.6 Components - The Central Processor Unit shall be comprised of pluggable, printed circuit modules containing integrated circuit packages, and a minimum amount of discrete network elements (Type 1, Type 2, and Type 5 or Type 1, Type 2, Type 6 and Type 7 maintenance modules).

3.5.1.7 Absolute Page Address Registers (APR) - Addressing for memory greater than 32,768 words is accomplished by the use of Absoltue Page Registers, in a translation table, to translate a 15-bit address into a 17-bit address for a maximum of 131,072 word memory. The page size will be 2048 words. The amount of memory to be addressed at any one time is 32,768 words.

Sixteen APRs will be used, each APR 7 bits long. Four bits of the 15-bit address (Y or P) are used to specify one of the 16 APRs, 6 bits of the APR plus the remaining 11 bits of the address word are concatenated to produce a 17-bit address (see Figure 6). The remaining 1 bit of the APR will indicate if the memory location is in a memory-protect mode.

3.5.2 Memory Unit - The Memory Unit shall meet the following requirements:

3.5.2.1 Function - The memory shall be capable of being operated in the overlapped mode. Each overlapped segment shall be a minimum of 16K, thirty-bit destructive readout (DRO) words or a multiple of 16K. To provide the overlapped mode shall require a minimum of two units of 16K, thirty-bit DRO words each.

3.5.2.2 Form Factor - Each Memory Unit shall be enclosed in a case 5.5 inches wide by 12.6 inches high by 14.4 inches long (handles and alignment pins excluded).

*3.5.2.3 Weight - The weight of each Memory Unit shall not exceed 35.5 pounds.

3.5.2.4 DRO Memory - Memory Units shall be composed of up to four 4,096 word DRO core memory stacks and the associated electronics. Memory shall be expandable in 8,192 word increments.

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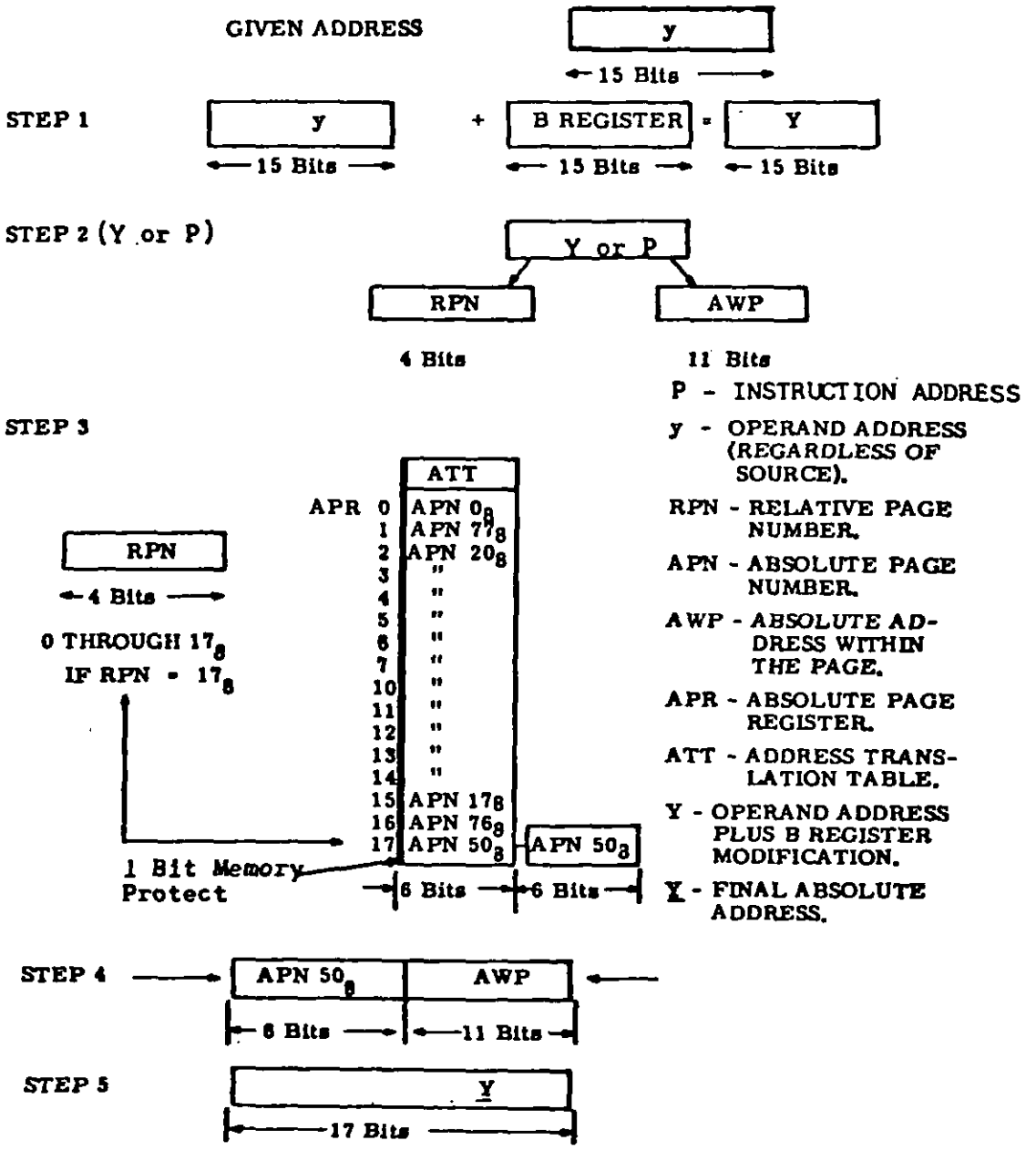


Figure 6. Indirect Addressing

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*3.5.2.5 Speed - The memory cycle time shall be two microseconds maximum. The memory cycle time shall be two microseconds ± 6 percent when measured with the central processor.

3.5.2.6 Components - Each unit shall consist of four core stacks (Type 3 Maintenance Module) of 4,096 words, thirty bits each, with the associated electronics mounted on pluggable Type 1 and 2 Maintenance Modules.

*3.5.2.7 Memory Allocations - Table IV specifies the memory allocations for the 65K memory version of the computer.

3.5.2.8 Memory Interconnections - The interconnections between the memory sense amplifiers and the memory stacks shall be through cables that are exclusive of the interconnection panel or shall be through the wire wrapped interconnection panel (see 1.2 and 6.6).

3.5.3 Input/Output Unit - The Input/Output Unit shall meet the following requirements.

*3.5.3.1 Function - The Input/Output Unit of the equipment shall provide computer control of, and communications with, peripheral equipment. The number of input channels will be either 12 or 16 (see 6.6). If only 12 input channels are used, the remaining 4 channels will be wired and connectors installed, but the Maintenance Modules will not be delivered. Each channel shall provide parallel data transfer for up to 30 data bits. Each group of 4 input channels and the corresponding 4 output channels shall have access to an assigned 32,768 word memory group, with selectability to other 32,768 word groups of a maximum 131,072 word memory, by making a backpanel wire change. The output configuration for this contract will have 16 output channels bussed into four output groups (Bussed Output Configuration). Each bussed output group will have two output connectors; 1 connector will provide a 30-bit parallel shared data path for the four output channels of the group and one set of control lines for the lowest priority channel of the group; the second connector will provide control signal communications for the three remaining output channels of the group.

*3.5.3.1.1 I/O Buffering Modes - The computer shall have the capability for communications with peripheral equipment on any and all I/O channels concurrently with the execution of the program. Buffering operations, once initiated by a programmed instruction, shall proceed to termination asynchronously with the program. The buffer instruction selects the I/O channel and designates the area of computer memory to be used by the channel for storing incoming data or dispatching outgoing data.

3.5.3.1.1.1 With Monitor - Buffering with the monitor shall transfer words sequentially, starting at a given initial address through a given terminal address, on the specified input or output channel. The computer shall continue execution of program instructions during the buffer process. Completion of the buffering process shall cause an internal interrupt transferring program control to the input, output or External Function monitor interrupt entrance register assigned to the input or output channel.

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TABLE IV.

Memory Allocations - 65,536 - Word Memory

Octal Address Range	Use
00000	Program Fault Interrupt Entrance Address
00001	Count-down Clock Interrupt Entrance Address
00002	Memory Protect Interrupt Entrance Address
00003	Input Power Failure Interrupt Entrance Address
00004	Power On Entrance Address
00005-00017	Unassigned
00020-00037	External Interrupt Entrance Address
00040-00057	Input Monitor Interrupt Entrance Addresses
00060-00077	Output Monitor Interrupt Entrance Address
00100-00117	Input Buffer Control Words
00120-00137	Output Buffer Control Words
00140-00157	External Function Buffer Control Words
00160	Real-Time Clock
00161	Count-down Clock
00162-00177	Unassigned
00200-00217	ESI Input Buffer Termination Words
00220-00237	ESI Output Buffer Termination Words
00240-00257	ESI External Function Buffer Termination Words
00260-00477	Unassigned
00500-00517	External Function Buffer Monitor Interrupt Entrance Addresses
00520-00537	Interrupt Word Storage Addresses
00540-00577	Unassigned
00600-00617	Intercomputer Time-Out Interrupt Entrance Address (one per channel)
000620-177777	Unassigned

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3.5.3.1.1.2 Without Monitor - Buffering without the monitor shall transfer words sequentially, starting at a given initial address through a given terminal address, on a specified input or output channel. The computer shall continue execution of program instructions during the buffer process. No monitor-interrupt shall occur upon termination.

3.5.3.1.2 Communication -

*3.5.3.1.2.1 Inputs from Peripheral Equipments - Inputs to the computer via the I/O Unit shall be made by either of two methods - Interrupt or Data.

3.5.3.1.2.1.1 Interrupt Inputs - A peripheral equipment shall be able to input one 30-bit word to the computer by following the control line sequence below.

a. Computer sets the Interrupt Enable line when it is ready to accept an External Interrupt.

b. The peripheral equipment detects the Interrupt Enable.

c. The peripheral equipment places the Interrupt word on the data lines.

d. The peripheral equipment sets the Interrupt line to indicate that the Interrupt word is on the data lines.

e. Computer detects the Interrupt signal and at its convenience, accepts the Interrupt word.

f. Computer drops the Interrupt Enable and sets the Input Acknowledge pulse.

g. The peripheral equipment detects the Input Acknowledge and the drop of the Interrupt Enable and clears the Interrupt line and the data lines.

The Input Acknowledge to an interrupt shall be initiated at the same time that the Interrupt Enable is cleared. The simultaneous occurrence of these conditions should be used by peripheral equipments to differentiate between an Interrupt Acknowledge and Data Acknowledge.

3.5.3.1.2.1.2 Data Inputs - A peripheral equipment shall be able to input one 30-bit word to the computer by the following sequence.

a. Computer initiates input buffer for given channel.

b. The peripheral equipment places a data word on the 30 data lines.

c. The peripheral equipment sets the Input Data Request line to indicate that it has data ready for transmission.

d. Computer detects the Input Data Request.

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e. Computer samples the 30 data lines at its convenience.

f. Computer sets the Input Acknowledge line, indicating that it has sampled the data.

g. The peripheral equipment senses the Input Acknowledge line.

h. The peripheral equipment drops the data lines and the Input Data Request Line.

Steps b through h of this sequence are repeated for every data word until the number of words specified in the input buffer has been transferred.

*3.5.3.1.2.2 Computer Outputs to Peripheral Equipments - Outputs from the computer can be made by either of two methods - Data or External Function.

3.5.3.1.2.2.1 Data Outputs - The computer systems shall be able, via the I/O Unit, to output one 30-bit word by following the control line sequence below.

a. Computer initiates output buffer for given channel.

b. The peripheral equipment sets the Output Data Request Line indicating that it is in a condition to accept data.

c. Computer detects Output Data Request.

d. Computer, at its convenience, places information on the 30 data lines.

e. Computer sets the Output Acknowledge line indicating that the data is ready for sampling.

f. The peripheral equipment senses the Output Acknowledge pulse and samples the 30 data lines.

g. Computer drops the Output Acknowledge line, the data lines may remain set.

h. The peripheral equipment drops the Output Data Request after detecting the Output Acknowledge.

Steps b through h of this sequence are repeated for every data word until the number of words specified in the output buffer have been transferred.

3.5.3.1.2.2.2 External Functions - The computer shall be able to output one 30-bit External Function Code by the following sequence.

a. The computer initiates an External Function buffer for a given channel.

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b. The peripheral equipment sets the External Function Request line indicating that it is in a condition to accept External Function codes.

c. Computer detects the External Function Request.

d. Computer, at its convenience, places the External Function code on the 30 data lines.

e. Computer sets the External Function line indicating that the code is ready for sampling.

f. The peripheral equipment senses the External Function pulse and samples the 30 data lines.

g. Computer drops the External Function line, the data lines may remain set.

h. The peripheral equipment drops the External Function Request after detecting the External Function.

Steps b through h of this sequence are repeated for each command word until the number of words specified in the external function buffer has been transferred. External Function Transfers will occur any time an external function buffer is active and the peripheral establishes an External Function Request. If an External Function with force is to be sent, then steps d, e, f, and g would form the sequence.

*3.5.3.1.2.3 Intercomputer Communications - Any or all of the computer I/O channels shall be capable of being used for intercomputer communication. This ability shall be accomplished by a wire change. The control signal lines for intercomputer communication are as indicated in Table V.

TABLE V. Intercomputer Communication Control Signal Lines.

Input Channel	a. Interrupt Enable b. External Interrupt c. Input Data Request d. Input Acknowledge
Output Channel	a. External Function Request b. External Function c. Ready d. Resume

3.5.3.1.2.3.1 Intercomputer Command Word Transfer - A command word can be transferred from Computer A to Computer B by the following sequence.

a. Computer A initiates an External Function buffer for the intercomputer channel.

b. Computer B sets the Interrupt Enable line indicating that it is in a condition to accept External Function codes from Computer A.

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c. Computer A detects the Interrupt Enable as an External Function Request.

d. Computer A places the External Function code on the 30 data lines, and sets the External Function line.

e. Computer B detects the External Function as an External Interrupt Enable line.

f. Computer B sets the Input Acknowledge line, and drops the Interrupt Enable line.

g. Computer A detects the Input Acknowledge as a resume and drops the External Function line.

Steps b through g of this sequence are repeated for each command word until the number of words specified in the external function buffer has been transferred.

3.5.3.1.2.3.2 Intercomputer Data Transfer - A data word can be transferred from Computer A to Computer B by the following sequence:

a. Computer B initiates an input buffer and Computer A initiates an output buffer on the intercomputer channel.

b. Computer A places data on the 30 data lines.

c. Computer A sets the Ready line to indicate that the data is on the line.

d. Computer B recognizes the Ready signal as an Input Data Request signal and at its convenience, accepts the data word.

e. Computer B sets the Input Acknowledge line.

f. Computer A recognizes the Input Acknowledge as a Resume signal and clears the Ready signal.

Steps b through f of this sequence are repeated for each word to be transferred.

*3.5.3.1.2.4 Externally Specified Index (ESI) Buffer Mode Transfer - Any input/output channel pair shall be capable of ESI buffer mode operation as an option. The modification necessary to convert the channels to ESI shall be capable of being accomplished at the installed site. ESI data transfer shall be identical with normal data transfer except that the computer requires that the external device specify the address of the Buffer Control Word (index). The index shall then control the data transfer. Any one ESI channel may specify any number of indexes (limited only by the memory addresses available) with each index controlling the data transfer for the subchannel. The ESI index memory addresses are from 00200-00377.

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3.5.3.1.2.5 ESI Input - The external device shall place the address of the sub-channel input index buffer Control Word in bits 07-00 of the input data word, the remaining bits (29-08) shall contain the data. An input data request shall be sent to the computer after the input data is sent on the input data lines. The request shall stay on until acknowledged by the computer. The computer shall store the entire 30-bit-input-data word at the current address specified by the index Buffer Control Word.

*3.5.3.1.2.6 ESI Output - The external device shall place the address of the subchannel output index Buffer Control Word in bits 07-00 on the input data lines. An output data request shall be sent to the computer, after the address has been set on the input data lines; the request shall remain set until the output request is acknowledged. Upon recognition of the output request, the computer will sample the input lines for the buffer address. The computer will read the content of the address specified by the index buffer control and will place the 30-bit word on the output data lines.

*3.5.3.1.2.7 ESI Subchannel Termination - Whenever an ESI subchannel buffer terminates (input, output or EF buffers), the computer shall store the address of the terminating subchannel index buffer control work index and input data word at the corresponding (input, output, or EF) ESI Buffer Terminate address for that input/output channel as indicated in Table VI.

TABLE VI. ESI Subchannel Termination.

Terminate Address	I/O Channel
00200-00217	Input Buffer Termination Addresses
00220-00237	Output Buffer Termination Addresses
00240-00257	EF Buffer Termination Addresses

*3.5.3.1.2.8 ESI Buffer Initiation and Use - The Input/Output instructions (f = 73, 74, 75 or 76) must be used to activate the input or output on the ESI channel. The ENTer and SToRe instructions are used to set the buffer limits in the assigned index buffer control word (not the ESI buffer terminate address) before execution of the I/O instruction to activiate the ESI channel. Upon termination of any index subchannel, the ESI channel will become inactive and it must be reinitiated to allow the other index buffer subchannels to proceed to completion. Care must be taken to reset the buffer limits on the terminated subchannel or to turn off the subchannel peripheral device. The buffer termination address (channel and function dependent) shall contain the index address for the subchannel terminated.

*3.5.3.1.2.9 I/O Priority - The channel priority shall be such that the higher the numerical value of the channel, the greater its priority. Therefore, in order of descending priority, the channels are (15, 14, 13, 12) (11, 10, 9, 8) (7, 6, 5, 4) (3, 2, 1 and 0). Parentheses indicate the channel groups.

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The pyramid of functional I/O priority is constructed of the following levels:

a. I/O Channel Group Priority (I/O Group 3 - Highest, I/O Group 0 - Lowest).

b. I/O Channel Priority within a group (Channel 3 - Highest, descending to Channel 0 - Lowest) (Group priority plus channel priority established channel 15 to 0 priority).

c. I/O Mode Priority for any channel (External interrupt input - Highest, External Function - 2nd, Output data 3rd, Input data - 4th).

The I/O Channel group priority may be assigned priorities 1-4 in any order, and the four input/output channels associated with each channel group may be assigned priorities 1-4 in any order. Priority assignments shall be changeable at the installed unit by making internal wiring changes.

3.5.3.1.2.10 I/O Interrupt Priority - I/O interrupts shall be serviced based upon the availability of the memory containing the interrupt entrance addresses, interrupt lockout, processor sequencing, and the following:

3.5.3.1.2.10.1 Intercomputer time-out interrupt

a. I/O Group Priority (see 3.5.3.1.2.9).
b. I/O Channel Priority within a group (see 3.5.3.1.2.9).

3.5.3.1.2.10.2 Function Interrupts

a. I/O Group Priority (see 3.5.3.1.2.9).
b. I/O Channel Priority (see 3.5.3.1.2.9).
c. I/O Mode Priority (see 3.5.3.1.2.9).

*3.5.3.1.3 Timing Description - Control line timing shall be based on a nominal two megahertz main timing cycle utilizing a normal input or output channel and shall be as illustrated in Figures 7 through 10 and as described in the following:

*3.5.3.1.3.1 Data Computer Inputs from Peripheral Equipment

3.5.3.1.3.1.1 Input Data Request Signal - The Input Data Request Signal shall indicate to the computer that data has been placed on the 30 data lines. The Input Data Request must be maintained on the lines until an Input Acknowledge is received. There is no maximum limit on the duration of the Input Data Request since its value for any particular cycle is determined by the interaction with the computer program and the other input/output channels. The data lines must remain stable as long as the Input Data Request is set.

3.5.3.1.3.1.2 Input Acknowledge - The Input Acknowledge shall indicate to the peripheral equipment that its 30 data lines have been sampled. The Input Acknowledge signal shall be set for a fixed time interval. Upon sensing the Input Acknowledge, the Input Data Request may be dropped to the "zero" state at any time, but it must be dropped before another Input Data Request can be initiated. The time relationships are such that the peripheral equipment desiring to send data at a maximum rate

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couple legitimately reset the Input Data Request before the previous Acknowledge has been dropped to the "zero" state, however, the Input Acknowledge shall always be returned to the "zero" state before being reset to the "one" state. This shall not affect operation of the cycle since a delay will occur between successive settings of the Input Acknowledge.

*3.5.3.1.3.2 Interrupt Inputs to the Computer System

3.5.3.1.3.2.1 Interrupt Enable Line - The Interrupt Enable line from the computer system shall indicate to the peripheral equipment that the computer is now ready to accept an external interrupt except when the interrupt is locked out internally under program control.

3.5.3.1.3.2.2 Interrupt Line - The Interrupt line shall indicate to the computer system that the peripheral equipment has placed data on the 30 data lines. The computer shall then accept the Interrupt word at its convenience. The computer shall then simultaneously drop the Interrupt Enable and set the Input Acknowledge. The peripheral equipment can now drop the Interrupt signal and Interrupt word. The data lines must remain stable as long as the Interrupt line is in the set condition.

3.5.3.1.3.3 Normal Outputs and External Functions from the Computer

3.5.3.1.3.3.1 Output Request or External Function - The peripheral equipment must set the Output Request line or External Function Request line indicating that it is in a condition to accept a data or External Function word from the computer. There shall be a delay between the setting of the Output Data Request or External Function Request and the placing of the data on the line. Data lines will not necessarily be cleared to the "zero" state before being reset to the "one" state.

3.5.3.1.3.3.2 Output Acknowledge or External Function - The Output Acknowledge or External Function signal shall indicate to the peripheral equipment that its requested word is now present on the data lines which should now be sampled. The Output or External Function Request may be dropped to the "zero" state any time after detecting the Output Acknowledge or the External Function. The Output or External Function Request cannot be reset immediately to indicate readiness of the peripheral equipment to accept a second word because the computer will not recognize the second Output or External Function request unless a minimum time delay is allowed between the start of the dropping of the first Output or External Function Request and the start of the second Output or External Function Request. The timing shall allow peripheral equipment, which wishes to receive data from the computer at a maximum rate, to legitimately set the output or External Function Request to the "one" state for the second time before the first Output Acknowledge or External Function has dropped to the "zero" state. However, this shall not affect operation of the cycle since a delay will again exist between the setting of the second Output or External Function Request and the availability of the second word on the data lines.

3.5.3.1.3.4 External Functions with Force - External Functions with force are unique in that no request is sent by the peripheral equipment. The computer shall place the External Function Code on the 30 output data lines, and subsequently energize the External Function line. The External Function signal shall indicate to the peripheral equipment that a signal shall indicate to the peripheral equipment that an External Function word is present on the data lines, which should now be sampled. The External Function

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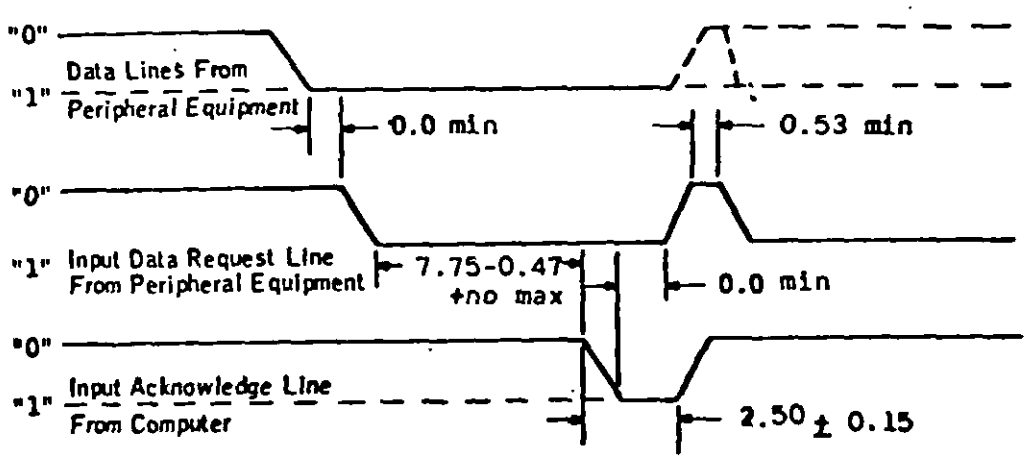


Figure 7. Data Inputs to Computer from Peripheral Equipment

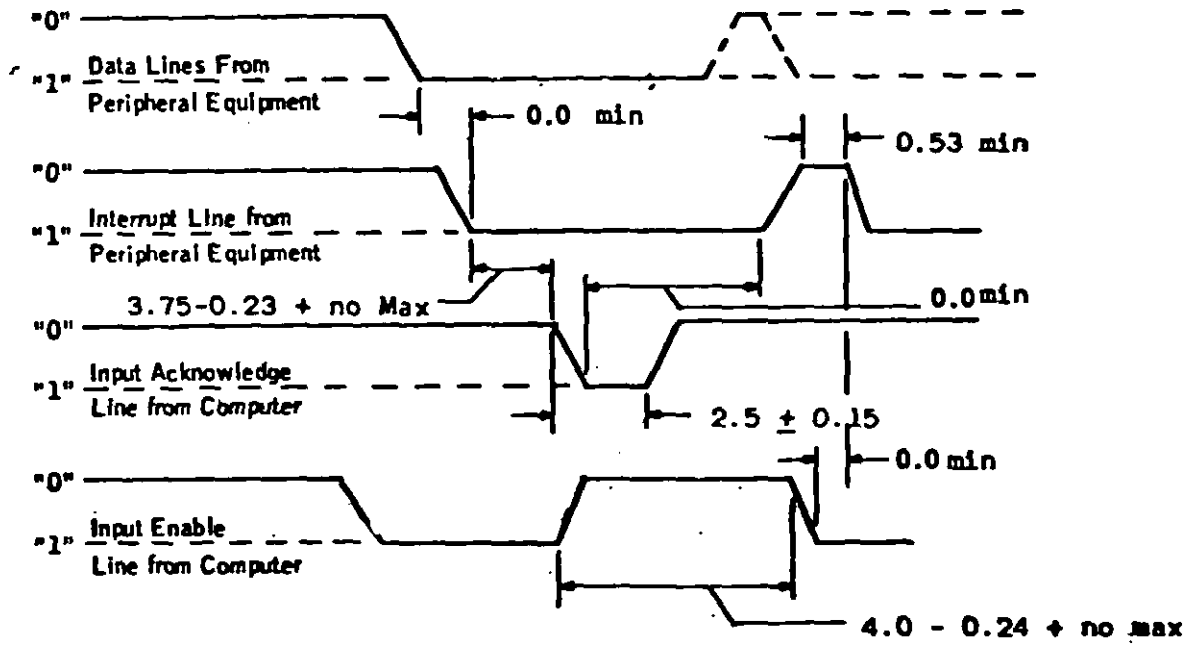


Figure 8. External Interrupt Inputs from Peripheral Equipment

NOTES:

1. All transition times are: Data Lines 1.0 usec, Control Lines 0.25 usec
2. All times are in microseconds
3. No max designation behind a time notation implies dependence on speed of peripheral equipment, I/O activity and program.

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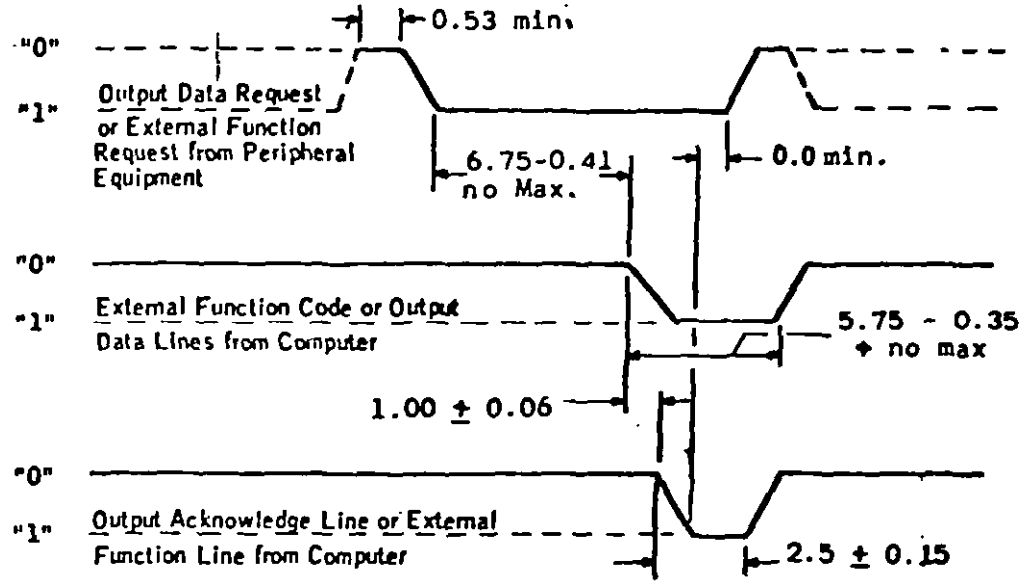


Figure 9. Data Outputs and External Functions from the Computer

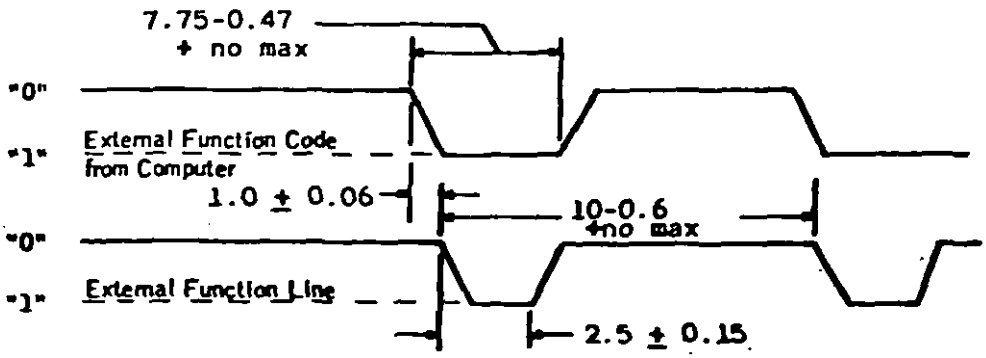


Figure 10. External Function with Force

NOTES:

1. All Transition times are: Data Lines 1.0 usec, Control Lines 0.25 usec .
2. All times are in microseconds
3. No max designation behind a time notation implies dependence on speed of peripheral equipment, I/O activity, and program.

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word shall remain on the data lines after the External Function signal begins to drop. The peripheral equipment has no control over the rate at which External Functions with force are sent. If the peripheral equipments cannot accept External Functions at the computer rate, restrictions must be made in the programming of the External Function instructions to the equipment. It should be noted that, if the peripheral equipments have no provisions for sending an External Function Request signal, all External Functions to existing peripheral equipment must be sent as External Function with force.

*3.5.3.2 Form Factor - The Input/Output Unit shall be enclosed in a case 11.0 inches wide by 10.6 inches high by 16.7 inches long. (Alignment pins and handles excluded).

*3.5.3.3 Weight - The weight of the Input/Output Unit shall not exceed 36 pounds including the tool box and tools.

*3.5.3.4 Data Word Transfer Rates - The Digital Computer Set shall be capable of transferring to peripheral equipment or receiving from peripheral equipment. 30-bit data words at the maximum rates as specified in Table VII. An I/O channel includes one input channel and one output channel.

TABLE VII. I/O Data Transfer Rates.

Number of Channels	Maximum Data Transfer Rates (± 6%) Input or Output Transfer Only	Input and Output Transfers Combined
One I/O Channel	100 KC	167 KC
One I/O Unit (4 I/O Channels)	100 KC	167 KC
Four I/O Units (16 I/O Channels)	167 KC	167 KC

3.5.3.5 Interface Signal Characteristics -

*3.5.3.5.1 Interface Logic Level Definitions - Following levels measured with cable meeting characteristics of 130 ohms impedance, bandwidth of 10 megacycles, attenuation of one precent or less.

*3.5.3.5.1.1 "One" Interface Logic Level - A logical "one" shall be represented by a potential difference across the output circuit terminals of 0.0 ± 0.5 volt.

3.5.3.5.1.2 "Zero" Interface Logic Level - A logical "zero" shall be represented by a potential difference across the output circuit terminals of +4.0 ± 1.0 volts.

3.5.3.5.1.3 Interface Threshold Level - The threshold level distinguishing the "one" state from the "zero" state is +1.5 ± 0.7 volts at the input amplifier terminal.

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3.5.3.6 Components - The I/O Unit shall be comprised of pluggable, printed circuit modules containing integrated circuit packages, and/or thin film hybrid circuit elements (Type 1 and/or 2 Maintenance Modules).

3.5.4 Power Supply Unit - The Power Supply Unit shall meet the following requirements:

3.5.4.1 Function - The power supply shall convert primary aircraft power to regulated voltages needed by the computer.

3.5.4.1.1 Control - On-Off control for the power supply shall be a remotely activated signal utilizing 28 VDC aircraft power.

3.5.4.1.2 Input Power - The power supply shall operate on input power as defined in 3.3.12.

*3.5.4.1.2.1 Transient Protection - In the event that the power supply cannot filter out severe power transients, such as encountered during generator switchover, sensing circuits shall be provided that will indicate to the Central Processor when an input transient below limit 3 of Figure 3 of MIL-STD-704 (issue specified in 2.1) as modified by 3.3.12 has been detected.

3.5.4.1.3 Output Power - The power supply shall supply the DC voltages required by the Central Processor, Input/Output Unit, Maintenance Console and up to a 65,536 word Memory System of the Digital Computer Set as described herein.

3.5.4.1.3.1 Regulation - Each output voltage shall remain within tolerance under the worst combinations of input voltage, input frequency and transients specified in 3.3.12 and temperature conditions as specified herein.

3.5.4.1.3.2 Protection - The power supply shall be designed to insure against excessive output over-voltage which might occur as a result of a component failure.

3.5.4.1.3.3 Overcurrent Protection - The power supply shall be designed to insure against excessive current drain from the supply due to a failure in the load.

3.5.4.1.3.4 Efficiency - The power supply shall have an efficiency of not less than 65 percent at an input of 117.5V line-to-neutral and maximum load per output.

*3.5.4.1.3.5 Power Factor - The power supply, power factor calculation shall be based on the ratio of the input power in watts to the input volt ampere values. The limits shall be as specified in paragraph 6.9.4 of MIL-STD-704.

3.5.4.1.4 Operational Safeguards - The power supply shall provide a Computer Hold signal and a Voltage Out-of-Tolerance signal to safeguard DRO memory and computer components and for automatic program recovery in the event of a power failure.

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*3.5.4.1.4.1 Power Failure - An Input Power Failure Interrupt signal shall be generated when an input transient below 80 VAC is detected. A Power Failure Interrupt shall occur at least one millisecond before the Voltage Out-of-Tolerance signal occurs. This interrupt signal will interrupt the normal program sequence and cause execution of the program sequence starting with the instruction in memory location 000 0003 (Reference Table III).

*3.5.4.1.4.2 Voltage Out-of-Tolerance/Voltage Restart - A Voltage Out-of-Tolerance signal shall be generated, within the power supply, when a critical DC operating voltage falls below the lower tolerance limit. When the Voltage-Out-of-Tolerance condition occurs, computer operation will halt until the DC operating voltage returns to normal. Upon return of the critical voltage to the normal operating level, the Out-of-Tolerance signal shall be removed. Return of the Out-of-Tolerance signal voltage to the normal operating level shall be interpreted as a restart signal to the computer, and shall cause a computer master clear and a Processor Restart at the Power On entrance address (000004).

3.5.4.1.4.3 Reserve Capacity - The power supply shall have enough reserve capacity to supply the output voltages within the normal operating tolerances for all the requirements described herein.

3.5.4.2 Form Factor - The Power Supply Unit shall be enclosed in a case 3.9 inches wide by 12.6 inches high by 14.4 inches long (alignment pins and handles excluded).

*3.5.4.3 Weight - The weight of the Power Supply Unit shall not exceed 32 pounds.

3.5.4.4 Components - The power supply shall be comprised of pluggable modules wherever practicable (Type 4 Maintenance Modules).

3.5.5 Mounting Rack - The Mounting Rack required to mount the computer in the operating environment shall meet the following requirements:

*3.5.5.1 Function - The Mounting Rack shall contain mounting slides, electrical connections and unit interconnection cabling and blowers for the 65,536 word computer units described herein. It shall provide mechanical protection for the computer units, except for the maintenance console, and a means of hard-mounting these to the airframe.

*3.5.5.2 Form Factor - The mounting rack shall be 51.0 inches high by 13.5 inches wide by 20 inches deep with the air filter assembly mounted (excluding cables, connectors, latches and/or standoffs and blower housing projections).

*3.5.5.3 Weight - The weight of the mounting rack shall not exceed 74 pounds including the air filter assembly.

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*3.5.5.4 Components - The Digital Computer Set components shall consist of the computer units, described in 3.5.1 through 3.5.4, and the associated connectors and cabling.

*3.5.5.5 Cooling Requirements - The Digital Computer Set shall require no more than 650 pounds of cooling air per hour.

*3.5.5.5.1 Cooling - For operation in the P-3C aircraft, the equipment shall provide blowers or other cooling means to insure satisfactory performance assuming ambient air temperature at 90°F at sea level and 85°F at 8000 feet. When used for other applications cooling requirements shall be determined by contractual agreement.

3.5.6 Maintenance Console - The Maintenance Console shall meet the requirements of a MIL-C-6781, Type II Console, unless otherwise specified herein.

3.5.6.1 Function - The Maintenance Console shall provide the man-machine interface for the Digital Computer Set. The console shall provide such controls, displays and logic as are necessary for controlling the operation of the computer and, in event of failure, to provide means for assisting in the isolation.

3.5.6.2 Form Factor - The Maintenance Console shall be 17.5 inches wide by 16.8 inches high by 5.5 inches deep (connectors, handles and mounting brackets excluded).

*3.5.6.3 Weight - The weight of the Maintenance Console shall not exceed 39 pounds.

*3.5.6.4 Controls and Indicators - The Maintenance Console shall contain, but not be limited to the types of controls and indicators shown in Figure 11 and described in Table V.

3.5.6.5 Remote Controls - The computer's control/indicate functions accessible via the Maintenance Console-Remote Console interface are:

(a) Power On-Off Control
(b) Bootstrap Auto Recovery/Manual Control
(c) Start-Step Control
(d) Over-Temperature Indicate
(e) Stop Control
(f) Run Indicate
(g) Stop Indicate

3.5.6.6 Components - The Maintenance Console shall be comprised of pluggable, printed circuit modules containing integrated circuit packages wherever practicable.

*3.5.6.7 Panel Lighting - The Maintenance Console Panel shall conform to requirements of MIL-C-6781, Type I Console, unless otherwise specified herein. The Maintenance Console shall be lighted by external means.

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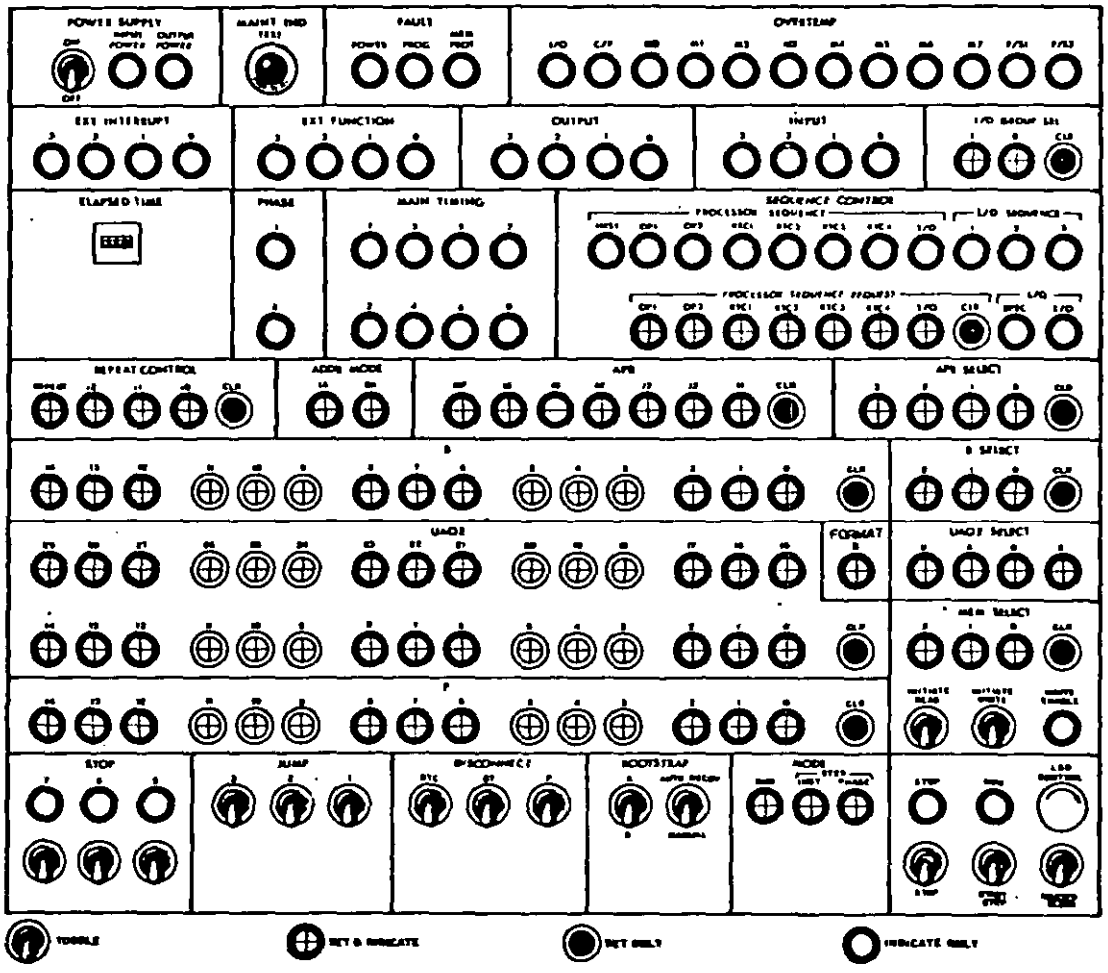


Figure 11. Maintenance Console

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TABLE VIII.

Maintenance Console Controls and Indicators

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
1		X	POWER SUPPLY ON/OFF	1A9S4	Toggle	1	Provides on-off control for the computer power supply.
	X		INPUT POWER	1A9S3	Neon-clear	1	Indicates 400 Hz power applied to power supply.
	X		OUTPUT POWER	1A9S2	Neon-clear	1	Indicates d-c power from power supply.
2		X	MAINT IND	1A9S1	Button-push	1	When activated, causes the panel's neon indicators to luminate.
3	X		FAULT POWER	1A9DS15	Incandescent-red	1	Indicates that a power fault has occurred.
	X		PROG	1A9DS14	Incandescent-red	1	Indicates that an illegal function code has been executed. The indication will be continuous until the computer is master cleared.
	X		MEM PROT	1A9DS13	Incandescent-red	1	Indicates that an instruction has been executed (in the indirect addressing mode) that initiated a memory write cycle into a protected memory area. The write is aborted under this condition.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
4	X		OVERTEMP	1A9DS1 thru 1A9DS12	Incandescent-red	12	Indicates which of the following units - input/output (I/O), central processor (C/P), memory (MO to M7) or power supply (P/S1 or P/S2) is experiencing an over-temperature condition.
5	X		EXT INTERRUPT	1A9S20 thru 1A9S23	Neon-clear	4	Indicates that one of the External Interrupt active flip-flops is set.
6	X		EXT FUNCTION	1A9S16 thru 1A9S19	Neon-clear	4	Indicates that one of the External Function active flip-flops is set.
7	X		OUTPUT	1A9S12 thru 1A9S15	Neon-clear	4	Indicates that one of the output active flip-flops is set.
8	X		INPUT	1A9S8 thru 1A9S11	Neon-Clear	4	Indicates that one of the input active flip-flops is set.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
9		X	I/O GROUP SEL*	1A9S6, thru 1A9S7	Button-push	2	When activated, each control (interpreted octal) will cause the corresponding I/O Group to be connected to the EXT INTERRUPT, EXT FUNCTION, OUTPUT and INPUT indicators.
	X				Neon-clear	2	Indicates which I/O group is connected to the OUTPUT, EXT FUNCTION, EXT INTERRUPT, and INPUT indicators.
10	X		ELAPSED TIME	1A9M1	Meter	1	Indicates the number of hours in tenths (up to total of four digits) during which power has been applied to the Maintenance Console.
11	X		PHASE	1A9S39, 1A9S54	Neon-clear	2	Indicates which one of the two clock phases is active.
12	X		MAIN TIMING	1A9S35 thru 1A9S38, 1A9S50 thru 1A9S53	Neon-clear	8	Indicates which of the main timing flip-flops are set.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
13			SEQUENCE CONTROL				
	X		PROCESSOR SEQ	1A9S27 thru 1A9S34	Neon-clear	8	Indicates whether a special I/O or processor sequence is enabled.
	X		I/O SEQUENCE	1A9S24 thru 1A9S26	Neon-clear	3	Indicates which I/O sequence is active.
		X	PROCESSOR SEQ REQUEST*	1A9S43 thru 1A9S49	Button-push	7	When activated, sets the processor sequences to be executed.
	X				Neon-clear	7	Indicates which processor sequences are to be enabled.
	X		L/O	1A9S40, 1A9S41	Neon-clear	2	Indicates whether a special or L/O interrupt has caused a lockout.
14			REPEAT CONTROL*	1A9S74	Button-push	1	When activated in conjunction with the OP1 PROCESSOR SEQUENCE REQUEST causes the instruction in U to be repeated.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
	X				Neon-clear	1	Indicates that the instruction in U will be repeated if the next sequence to be executed is an OPl sequence.
		X		1A9S71 thru 1A9S73	Button-push	3	When activated, each control causes a corresponding bit in the R-register to be set.
	X				Neon-clear	3	Indicates which of the three register bits are set.
15			ADDR MODE* IA/DA	1A9S68, 1A9S69	Button-push	2	The setting of this control selects either the direct addressing mode or the indirect addressing mode.
	X				Neon-clear	2	Indicates whether the direct or the indirect addressing mode has been selected.
16		X	APR*	1A9S61 thru 1A9S67	Button-push	7	When activated, the controls will cause corresponding bits of the selected APR to be set.
	X				Neon-clear	7	Indicates which of the seven bits of the selector APR are set.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
17		X	APR SELECT*	1A9S56 thru 1A9S59	Button-push	4	When activated, the controls will be interpreted octally and cause the corresponding APR to be connected to APR controls.
	X				Neon-clear	4	Indicates which APR is displayed in the APR indicators.
18		X	B*	1A9S80 thru 1A9S94	Button-push	15	When activated, each control causes a corresponding bit of the selected B-register to be set.
	X				Neon-clear	15	Indicates which of the bits of the selected B-register are set.
19		X	B SELECT*	1A9S76 thru 1A9S78	Button-push	3	When activated, the controls will be interpreted octally and cause the corresponding B-register to be connected to the B controls.
	X				Neon-clear	3	Indicates which of the index registers is in the B-register indicator.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
20		X	UAQZ*	1A9S100 thru 1A9S114, 1A9S120 thru 1A9S134	Button-push	30	When activated, each control causes a corresponding bit of the selected register to be set.
	X				Neon-clear	30	Indicates which of the 30 bits of the selected register are set.
21		X	FORMAT II*	1A9S99	Button-push	1	When activated, sets the Format II flip-flop.
	X				Neon-clear	1	Indicates that the instruction in the U-register is a Format II instruction.
22		X	UAQZ SELECT*	1A9S95 thru 1A9S98	Button-push	4	When activated, each control will cause the corresponding register to be connected to the UAQZ controls.
	X				Neon-clear	4	Indicates which of the registers is displayed in the UAQZ-register indicator

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
23		X	P*	1A9S138 thru 1A9S152	Button-push	15	When selected, each separate control causes a corresponding bit in the P-register to be set.
	X				Neon-clear	15	Indicates which of the 15 P-register bits are set.
24		X	MEM SELECT*	1A9S116 thru 1A9S118	Button-push	3	When activated, selects the Z-register that may be presented in the UAQZ-register (see note 1).
	X				Neon-clear	3	Indicates the memory bank being presented in the UAQZ-register.
		X	INITIATE READ	1A9S136	Toggle	1	When activated, initiates a manual read operation of memory address selected in P.
		X	INITIATE WRITE	1A9S135	Toggle	1	When activated, initiates a manual write operation to the memory address selected in P.
	X		WRITE ENABLE	1A9DS16	Incandescent-amber	1	Indicates that a manual read has been performed and that a manual write must be performed.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
25		X	STOP 7	1A9S172	Toggle	1	When activated, causes the computer to cease program operation if a function code of 61 or 65 with a j designator of 7 is executed.
	X		7	1A8S166	Neon-clear	1	Indicates that the computer has read a program instruction that had a function code of 61 or 65 and a j designator of 7 with STOP switch "7" set.
		X	6	1A9S171	Toggle	1	When activated, causes the computer to cease program operation if a function code of 61 or 65 with a j designator of 6 is executed.
	X		6	1A9S165	Neon-clear		Indicates that the computer has read a program instruction that had a function code of 61 or 65 and a j designator of 6 with STOP switch "6" set.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
		X	5	1A9S170	Toggle	1	When activated, causes the computer to cease program operation if a function code of 61 or 65 with a j designator of 5 is executed.
	X		5	1A9S164	Neon-clear	1	Indicates that the computer has read a program instruction that had a function code of 61 or 65 and a j designator of 5 with STOP switch 5 set.
26		X	JUMP 3	1A9S163	Toggle	1	When activated, causes the computer to jump to a special routine if an instruction that has a function code of 61 or 65 and a j designator of 3 is executed.
		X	2	1A9S162	Toggle	1	When activated, causes the computer to jump to a special routine if an instruction that has a function code of 61 or 65 and a j designator of 2 is executed.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
		X	1	1A9S161	Toggle	1	When activated, causes the computer to jump to a special routine if an instruction that has a function code of 61 or 65 and a j designator of 1 is executed.
27		X	DISCONNECT RTC	1A9S160	Toggle	1	When activated, causes the real-time-clock request to become disabled.
		X	B7	1A9S159	Toggle	1	When activated, prevents the B7 register from being automatically decremented.
		X	P	1A9S158	Toggle	1	When activated, prevents the incrementing of the P-register.
28		X	BOOTSTRAP A/B	1A9S157	Toggle-three position	1	Controls the three forcibly set entrance addresses to the bootstrap.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
		X	AUTO RECOVERY/ MANUAL	1A9S156	Toggle-three position	1	Controls the three conditions under which the bootstrap mode is enabled (center position inhibits the bootstrap mode).
29			MODE*				
		X	RUN	1A9S155	Button-push	1	When activated, enables the computer to execute one instruction when the START/STEP control is activated.
	X		RUN	1A9S155	Neon-clear	1	Indicates that the computer is prepared to operate in the run mode.
		X	STEP INST	1A9S154	Button-push	1	When activated, enables the computer to execute one instruction when the START/STEP control is activated.
	X		STEP INST	1A9S154	Neon-clear	1	Indicates that the computer is prepared in the instruction step mode.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
		X	STEP PHASE	1A9S153	Button-push	1	When activated, enables the generation of a single clock phase which advances through the instruction when the START/STEP is activated.
	X		STEP PHASE	1A9S153	Neon-clear	1	Indicates that the computer is prepared to operate in the phase step mode.
30		X	STOP	1A9S169	Toggle	1	When activated, causes the computer to cease program operation after completion of the next I sequence.
	X			1A9DS18	Incandescent-red	1	Indicates that computer operation has ceased.
31	X		RUN	1A9DS17	Incandescent-green	1	Indicates that the computer is in the run mode and is performing programmed instructions at normal speed or that the computer is operating in the phase step mode.

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TABLE VIII. Maintenance Console Controls and Indicators (Continued)

Item No.	Indicator	Control	Name/Description	Reference Designation	Type	Quantity	Function/Purpose
32		X	MASTER CLEAR	1A9S167	Toggle	1	When activated, causes the computer to be master cleared unless it is operating in the run mode. (see note 2.)
33		X	LSO CONTROL	1A9R1	Rheostat	1	Varies the output frequency of the low speed oscillator.
34		X	LSO START/STEP	1A9S168	Toggle-three position	1	When activated down, causes the computer to operate in the mode selected by the MODE control. When activated up, causes automatic START/STEP signal generation under the control and at the frequency of the low speed oscillator.

*The control and indicator are a single unit.

NOTES:

1. There is a Z-register for each of the 16K memory units.
2. There is a clear control for each of the following controls: UAQZ (1A9S119), B (1A9S79), B SELECT (1A9S75), P (1A9S137), I/O GROUP SEL (1A9S5), APR (1A9S60), APR SELECT (1A9S55), MEM SELECT (1A9S115), PROCESSOR SEQUENCE REQUEST (1A9S42) and REPEAT CONTROL (1A9S70).

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*3.5.7 Remote Control Panel - The Remote Control Panel shall conform to the requirements of MIL-C-6781, Type I Console, unless otherwise specified herein. The Remote Control Panel shall be subjected to separate contract negotiations and is not supplied under the existing contract(s).

3.5.7.1 Function - The Remote Control Panel shall be console mounted and provide controls for the following functions: Power On-Off, Start/Auto Recovery, and Stop/Bootstrap Manual. In addition, an indicator shall be provided for the following: Run, Stop and Overtemperature to indicate to the operator the status of the computer.

3.5.7.2 Form Factor - The Remote Control Panel shall be 5.75 inches by 2.25 inches high by 3 inches deep (connectors excluded).

3.5.7.3 Weight - The weight of the Remote Control Panel shall not exceed one (1) pound.

3.5.7.4 Edge Lit Panel - The Edge Lit Panel shall conform to applicable requirements of MIL-C-6781.

*3.5.8 Isolation Power Transformer Assembly - The Isolation Power Transformer Assembly shall meet the following requirements.

*3.5.8.1 Function - The Isolation Power Transformer Assembly shall be a separately mounted assembly and shall provide the input power coupling to the computer. The transformer shall provide attenuation of conducted electromagnetic interference generated by the computer.

*3.5.8.2 Form Factor - The Isolation Power Transformer Assembly shall be enclosed in a case 8.75 inches wide by 4 inches high by 14 inches long.

*3.5.8.3 Weight - The weight of the Isolation Power Transformer Assembly shall not exceed 26 pounds.

3.5.9 Configuration Management - Digital Computer Set configuration management shall be established to provide and maintain configuration control. Changes to the equipment shall be prepared in accordance with DOD-STD-480A.

3.5.9.1 Equipment Baseline - The equipment baseline shall be established after the completion of the Preproduction Tests performed on the selected preproduction unit. Modification or changes to the equipment as deemed necessary as a result of the Reliability Assurance Tests or the Maintainability Demonstration Test shall be incorporated only with the approval of the Procuring Activity.

3.5.9.2 Software Baseline - The computer diagnostics program baseline will be established upon satisfactory completion of the Maintainability Demonstration Test in accordance with the approved test plan. The computer Acceptance Test program baseline will be established after successful completion of the Acceptance Test performed on the first production equipment delivered.

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3.5.9.3 Integrated Circuitry Configuration Control - Prior approval of the Procuring Activity must be received before changes can be made to the type of integrated circuits used in the equipment.

*3.6 Growth Requirements - To provide assurance that future equipment requirements will be available, the growth provisions listed below shall be considered as computer options. These options when incorporated shall require a minimum of computer logic change and the options shall allow for field retrofit, when practicable.

*3.6.1 Memory Expansion - This option shall allow expansion to at least 131,072 thirty (30)-bit words of memory (see 3.5.1.5.3 for future information).

*3.6.2 Control Memory - This option shall provide control memory to contain input/output control information. When added to the computer, the control memory shall allow I/O rate of 167,000 thirty (30)-bit words per second on a single channel or 500,000 thirty (30)-bit words per second on three or more I/O units.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suit able for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services to conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by the specification shall be subjected to the following tests to determine compliance with all applicable requirements:

a. Preproduction tests, contract identified as First Article Approval Contractor Test.
b. Initial Production Tests.
c. Acceptance Tests.

4.1.2 Exceptions to Tests - Unless otherwise specified in the contract or purchase order, the following tests shall not be applicable:

a. Paragraph 4.3 and subparagraphs.
b. Paragraph 4.4.2 and subparagraphs.
c. Paragraph 4.4.3 and subparagraphs except as indicated in replacement paragraph 4.4.3.2 of this attachment.
d. Paragraph 4.4.4 and subparagraphs.
e. Paragraph 4.5 and subparagraphs.

4.2 Preproduction Tests - Preproduction tests shall be made on equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the

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the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government Inspector and the Procuring Activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having the adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Procuring Activity.

4.2.1 Preproduction Test Data - The contractor shall make available all data collected in conducting these tests to the Procuring Activity for review and approval. This data shall include, but not be limited to, a list of all electrical and electronic parts giving their specified voltage, current, and temperature rating; the applied circuit voltage and current; ambient, and surface temperatures of temperature critical parts. The ambient and surface temperature shall be obtained under the extreme high temperature operating conditions.

*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all requirements of this specification and the contract. Preproduction tests shall include the following:

a. Environmental Tests in accordance with MIL-T-5422.

b. Interference Tests in general accordance with WR-101, Part I, Amendment I, with the following modifications:

4.2.1.1.1 - line 5 - change 30 KC to 20 KC and the last sentence to read, "The broadband 100 db above one microampere per 20 KC limit as stated above is restricted to the NM-40".

c. Shakedown Tests

d. Power Input Tests

4.2.2.1 Shakedown Tests - Each preproduction equipment shall be subjected to a shakedown test of 100 continuous hours of failure-free operation. The test shall be conducted in accordance with MIL-STD-781 (issue specified in 2.1), using test level E and 5.2.3.2 except that the cycle time shall be eight (8) hours. Failures encountered in the test will be the subject of detailed failure analysis and corrective action. The corrective action loop shall document the assignable cause of failure and action taken. Corrective action shall be approved by the Procuring Activity prior to implementation. The test will then be started from time zero. The performance characteristics to be measured shall consist of a series of functional tests, exercising all portions of the equipment in a manner at least as stringent as required during its ultimate application.

a. Command, Arithmetic and Control Test - This test shall exercise and test each instruction and test all transmission paths within the arithmetic and control section.

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b. Memory Tests - These tests exercise all section of the memory sequentially by writing in the worst pattern of bits, circulating it through each memory stack, and verifying its contents with each shift.

c. Input/Output Tests - These tests shall exercise the input/output section as data transmission buffer operation, buffer termination, externally and internally generated interrupts.

4.2.2.2 Power Input Test - Each equipment shall be subjected to a range of input voltage and frequencies as specified herein.

a. The input voltage and frequencies shall be varied within the limits defined in 3.3.12. The equipment shall perform as specified in 3.5.

b. It shall be demonstrated that input voltage transients greater than those specified above shall not cause damage to the equipment and when the voltage returns within the specified limits normal operation shall automatically resume (see 3.5.4.1.4).

*4.2.3 Preproduction Approval - Approval of the Preproduction equipment shall be by the Procuring Activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample, except as authorized by the procuring activity. Prefabrication of production equipment prior to the approval of the preproduction sample shall be at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered to be one of the equipments under the contract.

*4.2.4 Production Equipment - Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each production equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. All tests such as burn-in or screening tests performed on components, assemblies and subassemblies, used in the preproduction equipments shall be continued for components, assemblies and subassemblies, used in production equipments. Evidence of noncompliance with the above shall constitute cause for rejection by the Procuring Activity. It shall be the obligation of the contractor to make necessary corrections as directed by the Procuring Activity.

*4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent to a designated Government laboratory for tests. This equipment shall be selected by the Procuring Activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test (see 4.4.1).

4.3.1 Scope of Tests - This equipment shall be subjected to any and all tests the Procuring Activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample and that it meets all applicable requirements including design, construction, workmanship, performance, and quality.

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4.3.2 Accessory Material - In addition to the complete equipment for Initial Production Tests, the contractor shall also make available such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the Procuring Activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the Procuring Activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when directed by the Procuring Activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the Procuring Activity shall engage the service of a commercial testing laboratory acceptable to the Procuring Activity. The contractor shall make available test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests
b. Sampling Tests
c. Reliability Assurance Tests
d. Special Tests.

*4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. Complete individual equipment acceptance shall constitute the satisfactory completion of the following tests:

a. Examination of Product
b. Operational Test
c. Manufacturing Run-In Test
d. All Equipment Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Tests - Each equipment shall be subjected, as a minimum, to the following operational tests after the equipment has been operated long enough to permit the equipment temperature to stabilize.

4.4.1.2.1 Manual Operations Tests - Each equipment shall be subjected to a manual operations test which demonstrates the functional ability of the controls and indicators to affect the equipment in the required manner.

4.4.1.2.2 Power Input Test - Each equipment shall be subjected to a range of input voltages and frequencies as specified herein.

a. The input voltage and frequencies shall be varied within the limits as specified in 3.3.12. The test program shall be a combination of those in 4.2.2.1 (paragraph a, b, and c).

b. It shall also be demonstrated that input voltage variations greater than those specified in 3.3.12 shall cause an automatic shut-down of the equipment and an automatic restart when the voltage returns within normal limits.

c. It shall also be shown that no information has been lost from memory after the above tests and normal operation can be resumed.

4.4.1.2.3 Functional Test - Functional testing shall be performed on each equipment to assure meeting the functional requirements of the specification.

*4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of six (6) hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of $\pm$ 2 g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten (10) minutes prior to the beginning of the six (6) hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration shall be normal to the plane of the printed circuit cards for ten (10) minutes. During the six (6) hour period of operation following ten (10) minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. The input power voltage and frequency shall also be cycled periodically between the specified tolerances. The input voltage

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transient test specified in 4.2.2.2 shall be conducted. Should a failure occur, it should be repaired and the test started over, except that the ten (10) minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The six (6) hour period specified above may be composed of two three-hour periods to conform with standard working hours.

*4.4.1.4 All Equipment Tests - Each of the production equipments submitted for acceptance testing shall be subjected to the All Equipment test specified in 4.4.3.2.1.

4.4.2 Sampling Tests - Sampling tests shall not be conducted unless the Reliability Assurance Test specified in 4.4.3 is deleted by contractual action. Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector as specified in Table IX.

TABLE IX. Sampling Tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be used on results of the preproduction, initial production, individual and special tests.

c. Manufacturing Run-In Test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action taken.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Tests required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested in accordance with MIL-STD-781, under the section entitled, "Qualification Phase of

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Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table V of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 All Equipment Test - As part of the acceptance test, each equipment produced shall perform six temperature start cycles in an alternating hot and cold temperature sequence. During the computer operating portion of the hot and cold start cycles, the computer shall be vibrated for ten minutes of every hour at a 2 g acceleration rate. Each of the above described three hot and three cold start cycles shall be performed failure free. If a failure occurs, the contractor will take corrective action and the test continued by re-starting at the beginning of the particular cycle in which the failure occurred. The second and subsequent failures in A.E.T. will require one of the following actions:

(a) Corrective action (replacement or repair) and restart of A.E.T. at the first segment. The contractor with government concurrence shall determine the readiness of the computer for restart of A.E.T. The computer will complete six (6) segments of A.E.T. failure free.

(b) The Contractor or Government has the option to remove the equipment from A.E.T. In the event the contractor or the Government removes the equipment from A.E.T., following corrective action (replacement or repair) the equipment shall be returned to the Contractor's quality confidence test.

When the equipment can operate through four (4) consecutive segments of the contractor's quality confidence test failure free, it shall be resubmitted to the Government for Manufacturing Run-In Test (at the Government option) and A.E.T. The submission of the equipment to A.E.T. shall be considered as an original submission. Failures occurring prior to removal of the equipment from A.E.T. are non-chargeable in determining reject criteria of Test Plan XXIX of MIL-STD-781. In the event of two (2) or more failures during A.E.T., a summary report on the failures shall be submitted to the Government representative before shipment of the equipment. The computer will be mounted as a unit to permit air flow equivalent to the mounting in the aircraft.

The hot and cold start cycles are defined as follows:

4.4.3.2.1 Hot Start - With computer power off, the computer temperature will proceed to and remain at +55°C for two hours. At the end of the two hour period, computer power will be applied and computer operation will begin. Computer operation will continue for twelve hours. At the same time computer power is applied, the test chamber temperature shall be lowered to +25°C within fifteen minutes. The chamber temperature will remain at +25°C for the remainder of the twelve hour computer operating time. At the end of the twelve hour computer operating cycle, computer power shall be turned off. The "hot" start cycle shall be repeated for a total of three cycles. At the completion of each "hot" cycle, the "cold" start cycle shall begin.

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4.4.3.2.2 Cold Start - With computer power off, the computer temperature will proceed to and remain at -25°C for two hours. At the end of the two hour period, computer power will be applied and computer operation will begin and continue for twelve hours. At the same time computer power is applied, the test chamber temperature shall be raised to +25°C within fifteen minutes. The chamber temperature will remain at +25°C for the remainder of the twelve hour computer operating time. At the end of the twelve hour computer operating cycle, computer power shall be turned off. The "cold" start cycle shall be repeated for a total of three cycles.

4.4.3.3 Performance Characteristics - The performance characteristics to be measured are part of the test procedures to be available for procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Test and shall be as specified in 6.9.

4.4.3.4 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the tests:

4.4.3.4.1 Performance Characteristics - Whenever performance characteristics fall below the acceptance requirements (see 6.9) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail. A failure will not be relevant when one or more of the following conditions exist:

a. The failure symptom cannot be duplicated during the subsequent retest.

b. It is a dependent failure.

c. The failure is a result of damage caused by mishandling improper maintenance, or discrepant conditions in equipment installation.

d. Failure occurred as a result of the equipment being subjected to conditions beyond that specified in the item specification.

e. Failures classified as "chargeable" may be excluded by mutual agreement between the Government and the Contractor when:

(1) The Contractor either analytically or by test establishes the effectiveness of the corrective action (or change) to the satisfaction of the Government.

(2) The proposed corrective action or change is approved by the Government.

(3) The corrective action or change is scheduled for introduction and/or retrofit by the Contractor as approved by the Government. Approval of any change by the Government shall not protect any subsequent failure from classification.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling of reliability assurance tests.

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4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the Procuring Activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the Procuring Activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for approval by the Procuring Activity prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare for Procuring Activity approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future equipments.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Maintainability Demonstration - A maintainability demonstration shall be conducted in accordance with Method I, Test Plan (b1) of Table II of MIL-STD-471, as follows to verify that the requirements of 3.3.7.1.1 are satisfied. The blowers shall not be considered as failure items for the maintainability demonstration.

4.5.1 Test Conditions - One (1) equipment shall be used for this test. The contractor shall select any one of the equipments that is available. All dust covers shall be in place on the equipment prior to the initiation of the test. Before beginning the test, the equipment shall be checked to determine that it is operating normally. Test equipment, tools, spare parts, diagnostic programs, and documentation shall be available for the test.

4.5.2 Repair Actions - A repair action shall be as follows (see 3.3.7.1.1):

a. Fault Isolation
b. Repair
c. Equipment Checkout

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4.5.3 Simulated-Failure - The failure shall be randomly introduced, one at a time, throughout the equipment. The points of failures shall be approved by the Procuring Activity prior to the initiation of tests.

4.5.3.1 Inducing Failures - Single failures shall be induced into the equipment by substituting modules that are known to be defective or, where defective modules are not available, simulated failures will be induced by applying grounds at test jacks or on extended modules. Any failure induced for demonstration purposes will consist of a continuous malfunction.

4.5.4 Repair Time Measurements - The time required for the technician to perform each repair action, including the use of technical manuals and support equipment, shall be measured starting at the time the technician is made aware that a malfunction exists and ending at the time the equipment is restored to normal operations. Any delay, such as waiting for parts, obtaining test equipment or administrative delays shall be excluded.

4.6 Test Procedures - The procedures used for conducting of all tests required by this specification, shall be prepared by the contractor and made available to the Procuring Activity for review and approval. The right is reserved by the Procuring Activity to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to preproduction, reliability assurance or maintainability tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge, and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PACKAGING

*5.1 General - All major unit and parts of the equipment shall be preserved, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is

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not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with Appendix D of MIL-STD-794.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended Use - The equipment covered by this specification is intended to be used in naval patrol aircraft for navigation, weapon delivery and other specific military missions. This equipment will be interconnected with displays, magnetic tape equipment, analog-to-digital conversion equipment and other related airborne equipment.

6.2 Test Values - Normal and limiting values of performance data shall be determined at the standard conditions specified in 3.3.9. This data is to be used in testing the equipment at installation points for compliance with minimum acceptance standards of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Procuring Activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-Repairable Subassemblies - As a general rule, repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Type Designation - The parenthesis (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used in nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data

6.6.1 Procurement Requirements - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1).
c. The nomenclature of the computer set (see 1.2).

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6.7 Changes From Previous Issues - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous issue.

*6.8 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Type Designations	Applicable Paragraph
Central Processor Unit	3.5.1
Memory Unit	3.5.2
Input/Output Unit	3.5.3
Power Supply Unit	3.5.4
Mounting Rack	3.5.5
Test Set, Computer Maintenance Console	3.5.6
Remote Control Panel	3.5.7
Transformer, Power, Isolation	3.5.8

*6.9 Performance Characteristics - The following listed computer performance characteristics should be measured to determine that they are in accordance with the related portions of paragraph 3.5 and the associated subparagraphs.

a. Instruction Execution Accuracy - Each instruction (with all of its variations and modifiers) shall be tested to determine that it is executed in accordance with the specification.

b. Instruction Execution Speed - Each instruction (with all of its variations and modifiers) shall be tested to determine that it is performed within the specified time.

c. Power Consumption - Computer power consumption shall be measured to determine that it is within the specified limit.

d. Memory Storage Capability - The computer memory shall be tested to determine that it can accurately store and recall the total amount of information required by the specification.

e. Memory Storage Rates - The DRO portion of memory shall be tested to determine that data can be stored at the rate specified.

f. NDRO Memory Permanence - The NDRO bootstrap memory shall be tested to determine that stored data cannot be altered by any electrical means.

g. Primary Power Variations - The computer shall be tested to determine the limits of input power over which it will satisfactorily operate. These limits must equal or exceed the specification requirements.

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h. Input/Output Data Rates - All input and output channels shall be tested to determine that the data rates for both single channel and multiple channel operation are in accordance with the specification.

i. Input/Output Electrical Characteristics - All input and output circuits shall be tested to determine that they perform as required by the appropriate specification.

j. Input/Output Independent Operation - All input and output channels shall be tested to determine that they operate independent of and simultaneously with program operation under the conditions specified.

k. Overlap Operation - The computer shall be tested to determine that the overlap mode of operation satisfies the appropriate specification requirements.

l. ESI - The computer shall be tested to demonstrate satisfactorily the capabilities of Externally Specified Indexing.

m. Maintainability - The computer shall be tested to determine compliance with maintainability requirements as specified herein.

Preparing activity:
Navy - AS

(Project 7021-N003)

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APPENDIX A
REPERTOIRE OF INSTRUCTIONS

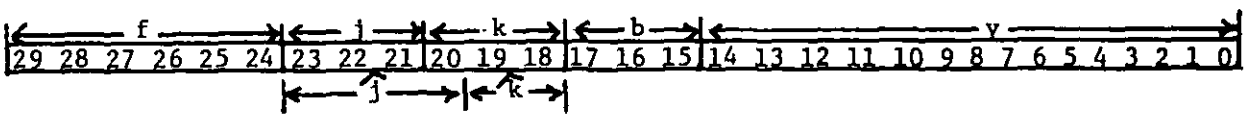
10. Scope.

10.1 Scope. The purpose of this appendix is to definitize the repertoire of instructions for the AN/ASQ-114 Computer Set.

20.0 The Avionics Unit Computer is a self-modifying, stored-program one-address computer. Although this means that one reference is provided for the execution of an instruction, this reference can be modified automatically during a programmed sequence. Instructions are in the form of 30-bit instruction words stored sequentially in memory; the computer executes the instructions sequentially unless an instruction alters the sequence.

20.1 By means of five designators, a computer instruction word completely defines a computer operation. The highest order six bits constitute the function code designator, f. The balance of the upper half of the instruction word contains the designators j, k, and b. The lower half of the word contains the operand designator y. The j and k are designators pertaining to input/output and Format II APR instructions.

AVIONICS UNIT COMPUTER INSTRUCTION WORD



20.2 FUNCTION CODE DESIGNATOR - f

The f designator appears in bit positions 29 through 24 of the U-register, or an instruction, designating the function to be performed by that instruction. Value 00_8 is a fault condition and, if executed, will cause a program fault interrupt.

20.3 BRANCH CONDITION DESIGNATOR - j

The j designator appears in bit positions 23, 22 and 21 of the U-register, or an instruction; it is used in a majority of the instructions for jump and skip determination, for B-register specification, and for repeat status interpretation. Appropriate interpretations of the j designator are listed below and under the descriptions of the individual instructions.

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For those instructions in which the j designator has no special interpretation, it specifies the condition under which the next sequential instruction in the program will be skipped. This permits branching from a sequence without executing a jump instruction, as would normally occur if a skip condition were not satisfied.

Skip of the next sequential instruction is determined by the following rules in all instructions except 04, 12, 13, 16, 17, 23, 26, 27, 40, 44, 60 through 76 and all Format II instructions:

j = 0: Execute the next instruction.
j = 1: Skip the next instruction.
j = 2: Skip the next instruction if (Q) is positive.
j = 3: Skip the next instruction if (Q) is negative.
j = 4: Skip the next instruction if (A) is zero (positive zero).
j = 5: Skip the next instruction if (A) is non-zero.
j = 6: Skip the next instruction if (A) is positive.
j = 7: Skip the next instruction if (A) is negative.

When the branch condition involves the sign of the quantity of A or Q, the evaluation examines the sign bit of these quantities: hence, positive zero is considered a positive quantity, and negative zero is considered a negative quantity. A negative zero, however, that is generated as a result of an arithmetic operation is normally forced to a positive zero.

NOTE: Bit position 29 is the sign bit; "1" = neg and "0" = pos.

20.4 INPUT/OUTPUT CHANNEL DESIGNATOR - $\hat{j}$

The $\hat{j}$ designator appears in bit positions 23, 22, 21 and 20 of the U-register of an input/output instruction, specifying the C-channel for the instruction. Thus the $\hat{j}$ provides accessibility to the 12 (decimal) input/16 (decimal) output channels numbered $0\text{-}17_8$. Bit positions 14, 13, 12 and 11 of the Format II APR instruction word, specify the APR for instruction. Instructions 13, 17, 62, 63, 66, 67, 73, 74, 75, 76, 7750 and 7754 use the $\hat{j}$ designator configuration.

20.5 OPERAND INTERPRETATION DESIGNATOR - k or $\hat{k}$

A k designator (3 bits) appears in bit position 20, 19 and 18 of the U-register, or an instruction; a k designator appears only in bit positions 19 or 18 (bit positions 10 and 9 for the Format II APR instruction) since bit 20 is a portion of the $\hat{j}$ designator. Instructions 13, 17, 62, 63, 66, 67, 73 through 76, 7750 and 7754 use the $\hat{k}$ designator configuration. The $\hat{k}$ designator controls both operand and instruction interpretation. The $\hat{k}$ designator is explained under the descriptions of the input/output related instructions.

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The k designator controls operand interpretation. Those instructions that read an operand, but do not replace it after the arithmetic is performed, are designated read instructions. Those instructions that do not read an operand, but store it, are designated store instructions. Instructions which both read and store operands are classified as replace instructions.

The various values of k affect the operands (except where noted otherwise under individual instruction description) as follows:

(a) Read instructions (01-12, 20-23, 26, 27, 30, 31, 40-43, 50-53, 60-65, 70-72, 7750):

k = 0: $¥_u$ = 0's; $¥_L$ = Y

k = 1: $¥_u$ = 0's; $¥_L$ = $(Y)_L$

k = 2: $¥_u$ = 0's; $¥_L$ = $(Y)_u$

k = 3: ¥ = (Y)

k = 4: $¥_u$ = All bits same as $¥_{14}$; $¥_L$ = Y

k = 5: $¥_u$ = All bits same as $¥_{14}$; $¥_L$ = $(Y)_L$

k = 6: $¥_u$ = Same bits as $¥_{29}$; $¥_L$ = $(Y)_u$

k = 7: ¥ = (A)

For instructions 22, 52, and 53, k = 7 is not used.

(b) Store instructions (14-16, 32, 33, 47, 7754):

k = 0: Store operand in Q*

k = 1: Store $operand_L$ in Y_L, leaving $(Y)_u$ undisturbed.

k = 2: Store $operand_L$ in Y_u, leaving $(Y)_L$ undisturbed.

k = 3: Store operand in Y.

k = 4: Store operand in the A-register.**

k = 5: Store complement of $operand_L$ in Y_L, leaving $(Y)_u$ undisturbed.

---

*A 1400000000 instruction complements (Q).

**A 1504000000 instruction complements (A).

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k = 6: Store complement of operand$_L$ in Y_u, leaving $(Y)_L$ undisturbed.

k = 7: Store the complement of operand in Y (storing the complement of B^j is the same as storing a 30-bit register complemented).

NOTE: Storing B^j, the B-register is treated as a 30-bit register, with the upper 15 bits set to 0. With a k = 7, store B^j will complement as a 30-bit register.

(c) Replace instructions (24, 25, 34-37, 44-46, 54-57):

k = 0: Not used.

k = 1: Read portion - $¥_u$ = 0's, $¥_L = (Y)_L$

Store portion - Store operand$_L$ in Y_L leaving $(Y)_u$ undisturbed

k = 2: Read portion - $¥_u$ = 0's; $¥_L = (Y)_u$

Store portion - Store operand$_L$ in Y_u leaving $(Y)_L$ undisturbed.

k = 3: Read portion - ¥ = (Y)

Store portion - Store operand in Y.

k = 4: Not used.

k = 5: Read portion - $¥_u$ = All bits same as $¥_{14}$; $¥_L = (Y)_L$.

Store portion - Store operand$_L$ in Y_L leaving $(Y)_u$. undisturbed.

k = 6: Read portion - $¥_u$ - All bits same as $¥_{29}$; $¥_L = (Y)_u$.

Store portion - Store operand$_L$ in Y_u leaving $(Y)_L$ undisturbed.

k = 7: Not used

NOTE: Repeat instructions require special interpretation when followed by a replace instruction.

20.6 INDEX DESIGNATOR - b

The b designator appears in bit positions 17, 16, 15 of the U-register, or an instruction and specified which of the B-index registers, if any, will be used to modify the operand address designator, y, to form $Y = y + (B^b)$.

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This operation employs and additive accumulator; hence, a quantity consisting of all zeros cannot result unless the bits of both the operand designator, y, and (B^b) are all zeros.

The effect of the various values of the b designator is a 15-bit one's complement sum as follows:

b = 0: Do not modify y.
b = 1: Add (B_1) to y.
b = 2: Add (B_2) to y.
b = 3: Add (B_3) to y.
b = 4: Add (B_4) to y.
b = 5: Add (B_5) to y.
*b = 6: Add (B_6) to y.
b = 7: Add (B^7) to y.

20.7 OPERAND DESIGNATOR - y

The y designator appears in bit positions 14 through 0 of an instruction. The operand or address of an operand, Y, is related to y through the expression Y = y + (B^b).

LEGEND

SYMBOL	REPRESENTS
$(A)_i$	Initial of A
$(A)_f$	Final content of A
(A)	Content of A
A_n	The nth bit of A
$(A)_n$	The nth bit of the content of A
f	Function code designator
j	Branch Condition designator
$\hat{j}$	Input/Output channel or APR designator
k or $\hat{k}$	Operand interpretation designator
b	Index designator
y	Operand designator
¥	Operand (regardless of source)

*Repeat instruction makes special use of B-register 6 when followed by a replace instruction.

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SYMBOL	REPRESENTS
Y	$y + (B^b)$ The operand or address for Read portion of an instruction. The destination address for the Store portion of an instruction.
(Y)	Content of memory address Y
L (Y) (Q)	Bit-by-bit multiplication, logical multiply of Y_n, and $(Q)_n$.
A	A-register or accumulator (30-bit arithmetic register)
B	Seven B-register (15 bits each). B-registers are address-modifying registers generally used for indexing loops in a program; in addition, B^7 serves as a repeat counter. (The address modification does not alter the instructions as stored in memory). The j or the b designator specifies the B-register used. The B-registers are located in the central processor.
Q	Q-register (30-bit arithmetic register)
U	U-register (30-bits). The U-register holds the instruction word during operand modification.
P	P-register (15-bits). The P-register is the program address register. This register holds the address of the next instruction throughout the program except for Jump instructions where the P-register is cleared and the new program address is entered.
C_j	The 16 input/16 output channels (30 data lines each). Channels consist of transmission lines; therefore, they cannot be considered registers. The designator j specifies (in octal) the channel used.

20.8 LIST OF INSTRUCTIONS

01 RIGHT SHIFT Q
This instruction causes an open-ended shift of (Q) to the right Y bit positions. The higher order bits are replaced with the original sign bit (bit position 29) as the word is shifted and the lower order bits are shifted out of Q. Only the lower order six bits of Y are recognized for this instruction, maximum shift count is 63 places.

02 RIGHT SHIFT A
This instruction causes an open-ended shift of (A) to the right Y bit positions. The higher order bits are replaced with the original sign bit (bit position 29) as the word is shifted out of A. Only the lower order six bits of Y are recognized for this instruction, maximum shift count is 63 places.

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03 RIGHT SHIFT AQ

This instruction causes an open-ended shift of (A) and (Q) as one 60-bit register to the right ¥ bit positions. The lower order bits of A are shifted into the higher order bit positions of Q and the lower order bits are shifted out of Q. The higher order bits of A are replaced with the original sign bit as the word is shifted. Only the lower order six bits of ¥ are recognized for this instruction, maximum shift count is 63 places.

04 COMPARE

This instruction compares the signed value of ¥ with the signed value of (A) and/or (Q). It does not alter either (A) or (Q). The branch condition designator, j, is interpreted in a special way for this instruction as listed below:

j = 0: Execute the next instruction

j = 1: Skip the next instruction

j = 2: Skip the next instruction if ¥ is less than, or equal to (Q).

j - 3: Skip the next instruction if ¥ is greater than (Q).

j = 4: Skip the next instruction if (Q) is greater than, or equal to ¥, and ¥ is greater than (A).

j = 5: Skip the next instruction if ¥ is greater then (Q), or ¥ is less than, or equal to (A).

j = 6: Skip the next instruction if ¥ is less than, or equal to (A).

j = 7: Skip the next instruction if ¥ is greater than (A).

05 LEFT SHIFT Q

This instruction shifts (Q) circularly to the left ¥ bit positions. The lower order bits are replaced with the higher order bits as the word is shifted. Only the lower order six bits of ¥ are recognized for this instruction. Maximum shift count permitted is 63 places.

06 LEFT SHIFT A

This instruction shifts (A) circularly to the left ¥ bit positions. The lower order bits are replaced with the higher order bits as the word is shifted. Only the lower order six bits of ¥ are recognized for this instruction. Maximum shift count permitted is 63 places.

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07 LEFT SHIFT AQ
This instruction shifts (A) and (Q) as one 60-bit register. The shift is circular to the left ¥ bit positions. The lower order bits of A are replaced with the higher order bits of Q, and the lower order bits of Q are replaced with the higher order bits of A. Only the lower order six bits of ¥ are recognized by this instruction. Maximum shift count permitted is 63 places.

10 ENTER Q
Transmit ¥ to Q.

11 ENTER A
Transmit ¥ to A.

12 ENTER Bj
Transmit ¥ to B-register j. The branch condition designator, j, is used to specify the selected B-register for this instruction and is not available for its normal function.

13 EXTERNAL FUNCTION ON $C\hat{j}$
This instruction establishes a 1-word external function buffer via the output buffer channel $\hat{j}$ from the storage address Y. The buffered word address is maintained on the lower order 15 bits of storage address 00140 plus $\hat{j}$.

This instruction is implemented as follows: For all $\hat{k}$ values, Y is stored in the upper and lower order half of storage location 00140 plus $\hat{j}$.

If $\hat{k}$ = 3, a 1-word external function buffer with force is established. (With force, the instruction ignores the external function request from the external device). The program will hold until the external function word is transmitted.

If $\hat{k}$ = 2, initiate a 1-word external function buffer and proceed to the next instruction. Subsequent to this instruction the transfer is accomplished when requested by the external device.

If $\hat{k}$ = 1, a 1-word external function buffer with monitor and with force is established. A monitor interrupt follows the completion of the buffering operations. Interrupt entrance address is 00500 plus $\hat{j}$.

If $\hat{k}$ = 0, a 1-word external function buffer with monitor is established. Proceed to the next instruction. A monitor interrupt follows the completion of the buffering operation. Subsequent to this instruction, the individual transfer is accomplished when requested by the external device. Interrupt entrance address is 00500 plus $\hat{j}$.

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14 STORE Q
Store (Q) at storage address Y as directed by the operand interpretation designator, k. If k = 0, complement (Q). If k = 4, store in A.

15 STORE A
Store (A) at storage address Y as directed by the operand interpretation designator, k. If k = 4, complement (A). If k = 0, store in Q.

16 STORE B^j
Store a 30-bit quantity, whose lower order 15 bits correspond to the content of B-register j and whose higher order 15 bits are zero, at storage address Y as directed by the operand interpretation designator k. The branch condition designator, j, is used to specify the selected B-register for this instruction and is not available for its normal function.

17 STORE $C^{\hat{j}}$ OR TEST EFB
If $\hat{k}$ = 3, transfer the interrupt word for storage address 00520 plus $\hat{j}$ to storage address Y.

If $\hat{k}$ = 2, transfer the contents of the C channel specified by $\hat{j}$ to storage address Y. An Input Data Acknowledge signal is then sent on the C channel. The program will hold until the word is read.

If $\hat{k}$ = 1, or 0, this instruction tests the status of the External Function (EF) buffer on the designated channel. If the buffer is active, a jump condition is satisfied. Then if $\hat{k}$ = 1, $(Y)_L$ or, if $\hat{k}$ = 0, Y becomes the address of the next instruction. If the buffer is inactive, the jump condition is not satisfied and the next sequential instruction in the current sequence is executed in the normal manner.

20 ADD A
Add ¥ to the previous content of the accumulator.

21 SUBTRACT A
Subtract ¥ from the previous content of the accumulator.

22 MULTIPLY
Multiply (Q) times ¥, leaving the double-length product in AQ. If the factors are considered as integers, the product is an integer in AQ. The actual multiplication is performed with positive numbers only; therefore, if the orignal sign bits of ¥ and Q are not similar, an end correction is made by complementing the product. The branch condition designator j is interpreted prior to the end correction; thus j = 7 has no effect and j = 6 always produces an unconditioned skip.

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23 DIVIDE/SQUARE ROOT
If k ≠ 7, divide (AQ) by Y leaving the quotient in the Q-register and the remainder in the A-register. If k = 7, take the square root of the (Q) leaving the root in the Q-register and the remainder in the A-register. The remainder a = n - (Q X Q), where n = the original number. The actual division is performed with positive numbers only; therefore, if the original sign bit of AQ is negative, the remainder is complemented. The branch condition designator is interpreted prior to the end correction; thus j = 7 has no effect and j = 6 always produces an unconditioned skip.

NOTE: If a divide overflow condition exists, no maintenance console indication is given. However, by coding each divide instruction with j = 3, a program test for the divide overflow is automatic. With this selection of j, a skip of the next instruction occurs if a divide overflow exists. The skip should be made to a jump instruction which provides a remedial means of noting the error or of correcting it. Therefore, the instruction which follows the divide instruction should have its j = 1 in order to preclude the the jump instruction whenever the divide sequence culminates in a correct answer.

A divide overflow can be detected also if the divide instruction is executed with j = 2. In this case, a correct answer is indicated when a skip occurs.

24 REPLACE A + Y
Add (Y) to the previous content of A. Store (A) at storage address Y.

25 REPLACE A - Y
Subtract (Y) from the previous content of A. Then store (A) at storage address Y.

26 ADD Q
Add ¥ to Q. The branch condition designator j has special meaning in this instruction as listed below.

27 SUBTRACT Q
Subtract ¥ from Q. The branch condition designator j has special meaning in this instruction as listed below.

In instruction 26 and 27 the branch condition designator j has the following meaning:

j = 0: Execute the next instruction.
j = 1: Skip the next instruction.
j = 2: Skip the next instruction if (A) is positive.
j = 3: Skip the next instruction if (A) is negative.
j = 4: Skip the next instruction if (Q) is zero.
j = 5: Skip the next instruction if (Q) is non-zero.
j = 6: Skip the next instruction if (Q) is positive.
j = 7: Skip the next instruction if (Q) is negative.

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30 ENTER ¥ + Q
Transmit ¥ + (Q) to A.

31 ENTER ¥ - Q
Transmit ¥ - (Q) to A.

32 STORE A + Q
Add (Q) to the previous content of A. Then store (A) at storage address Y as directed by the operand interpretation designator, k.

33 STORE A - Q
Subtract (Q) from the previous content of A. Then store (A) at storage address Y as directed by the operand interpretation designator, k.

34 REPLACE Y + Q
Transmit ¥ + (Q) to A. Store (A) at storage address Y.

35 REPLACE Y - Q
Transmit ¥ - (Q) to A. Store (A) at storage address Y.

36 REPLACE Y + 1
Transmit ¥ + 1 to A. Store (A) at storage address Y.

37 REPLACE Y - 1
Transmit ¥ - 1 to A. Store (A) at storage address Y.

40 ENTER LOGICAL PRODUCT
Enter the bit-by-bit product of ¥ and (Q) in A. The j designator is interpreted in a special way for this instruction for the values j = 2 or 3. If j = 2, skip if the parity of (A)f is even. If j = 3, skip if the parity of $(A)_f$ is odd. All other j values are normal.

NOTE: Even parity means an even number of "ones" in the A-register. Odd parity means an odd number of "ones" in the A-register.

41 ADD LOGICAL PRODUCT
Add to (A) the bit-by-bit product of ¥ and (Q).

42 SUBTRACT LOGICAL PRODUCT
Subtract from (A) the bit-by-bit product of ¥ and (Q).

43 COMPARE MASK
Subtract from (A) the bit-by-bit product of ¥ and (Q), and perform the branch point evaluation for skip of next sequential instruction as directed by the branch condition designator, j.

This instruction results in no net change in the content of any operational register. It provides, through the branch condition designator, j, a comparison of a portion of ¥ with (A).

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44 REPLACE LOGICAL PRODUCT
Enter in A the bit-by-bit product of (Y) and (Q). Then store (A) at storage address Y.

The j designator is interpreted in a special way for this instruction for the values j = 2 or 3. If j = 2, skip if the parity of $(A)_f$ is even. If j = 3, skip if the parity of $(A)_f$ is odd. All other j values are normal.

NOTE: Even parity - an even number of "ones" in the A-register.
Odd parity - an odd number of "ones" in the A-register.

45 REPLACE A + LOGICAL PRODUCT
Add to (A) the bit-by-bit product of (Y) and (Q). Then store (A) at storage address Y.

46 REPLACE A - LOGICAL PRODUCT
Subtract from (A) the bit-by-bit product of (Y) and (Q). Then store (A) at storage address Y.

47 STORE LOGICAL PRODUCT
Store in address Y the bit-by-bit product of (A) and (Q) as directed by the operand interpretation designator, k.

50 SELECTIVE SET
Set the individual bits of A to "one" corresponding to "ones" in ¥ leaving the remaining bits of A unaltered.

51 SELECTIVE COMPLEMENT
Complement the individual bits of A corresponding to "ones" in ¥, leaving the remaining bits of A unaltered.

52 SELECTIVE CLEAR
Clear the individual bits of A corresponding to "ones" in ¥, leaving the remaining bits of A unaltered. In this instruction k = 7, should not be used.

53 SELECTIVE SUBSTITUTE
Set the individual bits of A with bits of ¥ corresponding to "ones" in Q, leaving the remaining bits of A unaltered. In this instruction, k = 7 should not be used.

54 REPLACE SELECTIVE SET
Set the individual bits of A corresponding to "ones" in (Y), leaving the remaining bits of A unaltered. The store (A) at a storage address Y.

55 REPLACE SELECTIVE COMPLEMENT
Complement the individual bits of A corresponding to "ones" in (Y), leaving the remaining bits of A unaltered. Then store (A) at a storage address Y.

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56 REPLACE SELECTIVE CLEAR
Clear individual bits of A corresponding to "ones" in (Y), leaving the remaining bits of A unaltered. Then store (A) at storage address Y.

57 REPLACE SELECTIVE SUBSTITUTE
Clear individual bits of A corresponding to "ones" in Q, leaving the remaining bits of A unaltered. Then form the bit-by-bit product of (Y) and (Q), and set "ones" of this product in corresponding bits of A, leaving the remaining bits of A unaltered. Then store (A) at storage address Y.

60 JUMP (Arithmetic)
This instruction clears the program address register, P, and enters a new program address in P for certain conditions of either the A or Q-register content. The branch condition designator, j, is interpreted in a special way for this instruction and thus determines the conditions under which a jump in program address occurs. If the jump condition is not satisfied, the next sequential instruction in the current sequence is executed in a normal manner. If the jump sequence is satisfied, as listed below, ¥ defines the address of the next instruction and the beginning of a new program sequence. If k = 3 the computer mode will be set to the state specified by bit 2^{15} of the operand ¥.

j = 0: No jump. Reset interrupt enable and remove interrupt lockout, thus clearing bootstrap and interrupt modes. Continue with current program sequence.*

j = 1: Execute jump. Reset interrupt enable and remove interrupt lockout, thus clearing bootstrap and interrupt modes.

j = 2: Execute jump if (Q) is positive.

j = 3: Execute jump if (Q) is negative.

j = 4: Execute jump if (A) is zero.

j = 5: Execute jump if (A) is non-zero.

j = 6: Execute jump if (A) is positive.

j = 7: Execute jump if (A) is negative.

*NOTE: If both special and I/O interrupt lockouts are set, then the first f = 60, j = 0 + 1 will clear the special interrupt lockout and the second f = 60, j = 0 + 1 will clear the I/O interrupt lockout.

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61

JUMP (MANUAL)
This instruction clears the program address register, P, and enters a new program address in P for certain conditions of manual JUMP key selections. The branch condition designator, j, is interpreted in a special way for this instruction and thus determines the conditions under which a jump in program address occurs. If the jump condition is not satisfied, the next sequential instruction of the current sequence is executed in a normal manner. If the jump condition is satisfied, as listed below, ¥ defines the address of the next instruction and the beginning of a new program sequence. If k = 3, the computer mode will be set to the state specified by bit 2^{15} of the operand ¥.

Program execution may be stopped by certain STOP selections on execution of this instruction. The branch condition designator, j, specifies which key selections are effective.

j = 0: Execute jump regardless of key selections.

j = 1: Execute jump if JUMP 1 is selected.

j = 2: Execute jump if JUMP 2 is selected.

j = 3: Execute jump if JUMP 3 is selected.

j = 4: Execute jump. Stop execution.

j = 5: Execute jump. Stop execution if STOP 5 is selected.

j = 6: Execute jump. Stop execution if STOP 6 is selected.

j = 7: Execute jump. Stop execution if STOP 7 is selected.

62 JUMP ON $C^{\hat{j}}$ ACTIVE INPUT BUFFER
This instruction clears the program address register, P, and enters a new program address in P for certain input buffer conditions on the channel designated by $\hat{j}$. If the buffer is active, the jump condition is satisfied; then ¥ defines the address of the next instruction. If the buffer is inactive, the jump condition is not satisfied. The next sequential instruction in the current sequence is executed in the normal manner. k = 0, 1, 2, or 3 permitted.

63 JUMP ON $C^{\hat{j}}$ ACTIVE OUTPUT BUFFER
This instruction clears the program address register, P, and enters a new address in P for certain output buffer conditions on the channel designated by $\hat{j}$. If the buffer is active,

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the jump condition is satisfied, then ¥ defines the address of the next instruction. If the buffer is inactive, the jump condition is not satisfied. The next sequential instruction in the current sequence is executed in the normal manner. k = 0, 1, 2 or 3 permitted.

64 RETURN JUMP (Arithmetic)
This instruction shall execute a return-jump sequence for certain conditions of either the A or Q register content. The branch condition designator, j, and the address designator, k, are interpreted in a special way for this instruction. The j designator is used as a condition for execution of the return-jump and the k designator defines the address interpretation and that the addressing mode will be unchanged or set to Direct addressing mode. The return-jump will be satisified under the conditions listed below:

j = 0: No action. Continue with the current program sequence.

j = 1: Execute return jump.

j = 2: Execute return jump if (Q) is positive.

j = 3: Execute return jump if (Q) is negative.

j = 4: Execute return jump if (A) is zero.

j = 5: Execute return jump if (A) is non-zero.

j = 6: Execute return jump if (A) is positive.

j = 7: Execute return jump if (A) is negative.

If k = 0, 1, 2, 3, or 7 and the return jump is to be executed, store (P) in the lower half of memory address Y and the computer addressing mode (Direct or Page) in bit 2^{15} of ¥. The jump to ¥+1.

If k = 4, 5 or 6 and the return-jump is to be executed, set the addressing mode to Direct Addressing, store (P) in the lower half of memory address Y and the previous mode in bit 2^{15} of ¥. Then jump to ¥ + 1. Note the instruction and operand shall be read from memory before the addressing mode changes; the P store and jump shall be done utilizing the direct addressing mode.

If no jump is to be executed, the addressing mode shall remain unchanged.

65 RETURN JUMP
This instruction shall execute a return-jump sequence for certain conditions of manual key selections. The branch

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condition designator, j, and the address designator k are interpreted in a special way for this instruction. The j designator is used as a condition for execution of the return-jump and the k designator defines the address interpretation and that the addressing mode will be unchanged or set to Page addressing mode. The return-jump will be satisfied under control of the j designator conditions listed below:

j = 0: Execute return jump regardless of key selections.

j = 1: Execute return jump if JUMP 1 is selected.

j = 2: Execute return jump if JUMP 2 is selected.

j = 3: Execute return jump if JUMP 3 is selected.

j = 4: Execute return jump. Then stop execution.

j = 5: Execute return jump. Stop execution if STOP 5 is selected.

j = 6: Execute return jump. Stop execution if STOP 6 is selected.

j = 7: Execute return jump. Stop execution if STOP 7 is selected.

If k = 0, 1, 2, 3 or 7 and the return jump is to be executed store (P) in the lower half of memory address ¥ and the computer addressing mode (Direct or Page) in bit 2^{15} of ¥. Then jump to ¥ + 1.

If k = 4, 5, or 6 and the return-jump is to be executed, set the addressing mode to Page addressing, store (P) in the lower half of memory address ¥, and the previous mode in bit 2^{15} of ¥. Then jump to ¥ + 1. The instruction and operand shall be read from memory prior to setting the Page mode, and the (P) store, previous mode store and the jump shall be done in the Page mode.

If no jump is to be executed, the addressing mode shall remain unchanged.

66 TERMINATE $C^{\hat{j}}$ INPUT BUFFER OR ENABLE, DISABLE INTERRUPTS
If $\hat{k}$ = 0, terminate the input buffer on channel $\hat{j}$. No input buffer monitor interrupt will occur. If $\hat{k}$ = 1, b = 0, enable all interrupts. If $\hat{k}$ = 1, b ≠ 0, disable all interrupts. If $\hat{k}$ = 2, b = 0, enable all external interrupts. If $\hat{k}$ = 2, b ≠ 0, disable all external interrupts. If $\hat{k}$ = 3, b = 0,

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enable the external interrupt on the channel specified by $\hat{j}$. If $\hat{k} = 3$, $b \neq 0$, disable external interrupt on the channel specified by $\hat{j}$. The index designator, b, and the operand address designator, y, are not translated for this instruction.

67 TERMINATE $C^{\hat{j}}$ OUTPUT BUFFER, ALL BUFFERS, OR TERMINATE C^{n} EXTERNAL FUNCTION

If $\hat{k} = 0$, terminate the output data buffer on channel $\hat{j}$. If $\hat{k} = 1$, terminate the External Function Buffer on channel $\hat{j}$. If the channel specified by $\hat{j}$ is involved with the use of an output register of an intercomputer group, a resume signal is simulated and sent to the register. If $\hat{k} = 2$, terminate all buffers. If an output register is being used for intercomputer communications, a resume signal is simulated and sent to this register. For all values of $\hat{k}$, no output buffer monitor interrupt will occur. The index designator, b, and the operand address designator, y, are not translated for this instruction.

70 REPEAT

Transmit ¥ to B^7. If ¥ is non-zero, transmit (j) to r (repeat designator register), thereby initiating the repeat mode. If ¥ is zero, skip the next instruction.

The repeat mode executes the instruction immediately following the Repeat instruction ¥ times. B^7 contains the number of executions remaining throughout the repeat mode.

If no skip condition is met for the repeated instruction, the repeat mode will terminate when the repeat count in B^7 is exhausted and the instruction following the repeated insturction is executed. If the skip condition for the repeated instruction is met, the repeat mode terminates, and instruction following the repeated instruction is skipped. Following the repeat mode termination, the count remains in B^7.

In no way does the repeat mode alter a repeated instruction as stored in memory.

The three low-order bits of the r-designator (from j of instruction 70) affect the operand indexing as follows:

r = 0: Do not modify Y of the repeated instruction after each individual execution.

r = 1: Increase Y of the repeated instruction by one after each execution of the repeated instruction.

r = 2: Decrease Y of the repeated instruction by one after execution of the repeated instruction.

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r = 3: Repeat the initital B-register modification of the repeated instruction before each execution.

r = 4: Do not modify the operand address of the repeated instruction after each individual execution. If the repeated instruction is a replace instruction, the operand address is modified by (B^6) for the store portion of the replace instruction.

r = 5: Increase the operand address of the repeated instruction by one each execution of the repeated instruction. If the repeated instruction is a replace instruction, the incremented operand address is modified by (B^6) for the store portion of the replace instruction.

r = 6: Decrease the operand address of the repeated instruction by one after each execution of the repeated instruction. If the repeated instruction is a replace instruction, the decremented operand address is modified by (B^6) for the store portion of the replace instruction.

r = 7: Repeat the initial B-register modification of the repeated instruction before each execution. If the repeated instruction is a replace instruction, the modified operand address is further modified by (B^6) for the store portion of the replace instruction.

NOTE: Instruction 70 j designator establishes the repeat mode r designator. The 70, 71, and 72 instructions cannot be repeated.

71 B SKIP ON B^j

If the content of the B-register j is equal to ¥, skip the next instruction in the current sequence and proceed to the following instruction. Clear B-register j.

If the content of B-register j is not equal to ¥, proceed to the next instruction in the sequence in a normal manner. Increase the content of B-register j by one.

The branch condition designator, j, is used to designate the selected B-register in this instruction and is not available for its normal function. Only 15 bits of ¥ are used in the comparison described in the preceding paragraph.

72 B JUMP ON B^j

If the content of B-register j is non-zero, execute a jump in program address as defined by ¥. Reduce the content of B-register j by one.

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If the content of B-register j is zero, proceed to the next instruction in a normal manner. Do not alter the content of B-register j.

The branch condition designator, j, is used to designate the selected B-register in this instruction and is not available for its normal function. If the jump condition is satisified, then ¥ defines the address of the next instruction and the beginning of a new program sequence.

73 INPUT BUFFER ON $C^{\hat{j}}$ (without Monitor Mode)
This instruction establishes an input buffer via input channel $\hat{j}$ utilizing an area of memory defined by ¥. The memory address defined by ¥ is transferred to the buffer control register associated with the channel. Subsequent to this instruction, individual transfers will be executed at a rate determined by the external device. The storage address initially established by this instruction will be advanced by one following each individual transfer. The next current address will be maintained throughout the buffer process in the lower order 15 bits of control register 00100 plus $\hat{j}$. This mode will continue until it is superseded by a subsequent initiation or termination of an input buffer via the same input channel or until the higher order half and the lower order of control register 00100 plus $\hat{j}$ contain equal quantities, whichever occurs first. It should be noted that since the storage required for a buffering operation may be of any size up to 32,768 words utilizing any area of computer memory, it is a program responsibility to ensure that correct values for the buffer limits have been established for ¥ previous to the execution of this instruction.

This instruction is implemented as follows: If $\hat{k}$ = 3, store (Y) in storage location 00100 plus $\hat{j}$. If $\hat{k}$ = 1, store the lower order 15 bits of (Y) in the lower order half of storage location 00100 plus $\hat{j}$, leaving higher order half undisturbed. If $\hat{k}$ = 0, store Y in the lower order half of storage location 00100 plus $\hat{j}$ leaving the higher order half undisturbed. Proceed to the next instruction. $\hat{k}$ = 2 is not permitted.

74 OUTPUT BUFFER ON $C^{\hat{j}}$ (without Monitor Mode)
This instruction establishes an output buffer (if $\hat{k} \neq 2$, output data buffer; if $\hat{k}$ = 2, External Function buffer) via output channel $\hat{j}$ utilizing an area of memory defined by ¥. The memory address limits defined by ¥ are transferred to the buffer control register associated with the channel. Subsequent to this instruction, the individual transfers will be executed at a rate determined by the external device. The storage address initially established by this instruction will be advanced by one following each individaul transfer.

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The next current address will be maintained throughout the buffer process in the lower order 15 bits of the control register (if $\hat{k} \neq 2$, 00120 plus $\hat{j}$; if $\hat{k} = 2$, 00140 plus $\hat{j}$). This mode will continue until it is superseded by a subsequent initiation or termination of an output buffer via the same output channel or until the higher order half and the lower order half of the control register (if $\hat{k} \neq 2$, 00120 plus $\hat{j}$; if $\hat{k} = 2$, 00140 plus $\hat{j}$) contain equal quantitites, whichever occurs first. It should be noted that since the storage required for a buffering operation may be of any size up to 32,768 words, utilizing any area of computer memory, it is a program responsibility to ensure that correct values for the buffer limits have been established for ¥ previous to the execution of this instruction.

This instruction is implemented as follows: If $\hat{k} = 3$, store (Y) in storage location 00120 plus $\hat{j}$. If $\hat{k} = 1$, store the lower order 15 bits of (Y) in the lower order half of storage location 00120 plus $\hat{j}$, leaving the higher order half undisturbed. If $\hat{k} = 0$, store Y in the lower half of storage location 00120 plus $\hat{j}$, leaving the higher order half undisturbed. If $\hat{k} = 2$, store (Y) in storage location 00140 plus $\hat{j}$. Proceed to the next instruction.

75 INPUT BUFFER ON C^{j} (with Monitor Mode)
This instruction establishes an input buffer via input channel $\hat{j}$ utilizing an area of memory defined by ¥. The memory address limits defined by ¥ are transferred to the buffer control register associated with the channel. Subsequent to this instruction, the individual transfers will be executed at a rate determined by the external device. The storage address initially established by this instruction will be advanced by one following each individual transfer. The next current address will be maintained throughout the buffer process in the lower order 15 bits of control register 00100 plus $\hat{j}$. This mode will continue until it is superseded by a subsequnet initiation or termination of an input buffer via the same input channel or until the higher order half and the lower order half of the control register contain equal quantities, whichever occurs first. The inititaion of this input buffer selects the input channel $\hat{j}$ and establishes a buffer monitor on input channel $\hat{j}$. A monitor interrupt to address 00040 plus $\hat{j}$ follows completion of the buffering operation. It should be noted that since the storage required for a buffering operation may be of any size utilizing any area of computer memory up to 32,768 words, it is a program responsibility to ensure that correct values for the buffer limits have been established for ¥ previous to the execution of this instruction.

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This instruction is implemented as follows: If $\hat{k}$ = 3, store (Y) in storage location 00100 plus $\hat{j}$. If $\hat{k}$ = 1, store the lower order 15 bits of (Y) in the lower order half of storage location 00100 plus $\hat{j}$ leaving the higher order half undisturbed. If $\hat{k}$ = 0, store Y in the lower order half of storage location 00100 plus $\hat{j}$. Proceed to the next instruction. $\hat{k}$ = 2 is not permitted.

76 OUTPUT BUFFER ON C^j (with Monitor Mode)
This instruction establishes an output buffer (if $\hat{k} \neq 2$, output data buffer; if $\hat{k}$ - 2, external function buffer) via output channel j utilizing an area of memory defined by ¥. The memory address limits defined by ¥ are transferred to the buffer control register associated with the channel. Subsequent to this instruction, the individual transfers will be executed at a rate determined by the external device. The storage address initially established by this instruction will be advanced by one following each individual transfer. The next current address will be maintained throughout the buffer process in the lower order 15 bits of control register if $\hat{k}$ = 2, 00140 plus $\hat{j}$; if $\hat{k} \neq 2$, 120 plus $\hat{j}$. This mode will continue until it is superseded by a subsequent initiation or termination of an output buffer via the same output channel or until the higher order half and the lower order half of the control register contain equal quantities, whichever occurs first. The initiation of this output buffer selects the output channel $\hat{j}$ and establishes a buffer monitor on output channel $\hat{j}$. A monitor interrupt (if k ≠ 2, to address 00060 plus $\hat{j}$; if $\hat{k}$ = 2 to address 00500 plus $\hat{j}$) follows the completion of the buffering operation. It should be noted that since the storage required for a buffering operation may be of any size utilizing any area of computer memory up to 32,768 words, it is a program responsibility to ensure that correct values for the buffer limits have been established for ¥, previous to the execution of this instruction.

This instruction is implemented as follows: If $\hat{k}$ = 3, store (Y) in storage location 00120 plus $\hat{j}$. If $\hat{k}$ = 1, store the lower order 15 bits of (Y) in lower order half of storage location 00120 plus $\hat{j}$ leaving the higher order half undisturbed. If $\hat{k}$ = 2, store (Y) in storage location 00140 plus $\hat{j}$. Proceed to the next instruction.

77 11 ENTER A USING ADDRESS IN U
This instruction enters A with the contents of the address given in the 17 least significant bits of U. The address does not go through address translation; therefore, the instruction allows addressing of all memory without regard to the defined memory domain. This instruction utilizes the Type II General Format.

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77 15 STORE A USING ADDRESS IN U
This instruction stores (A) in address given by the 17 least significant bits of U. The address does not go through address translation; therefore, the instruction allows addressing of all memory without regard to the defined memory domain. This instruction utilizes the Type II General Format.

77 44 TEST AND SET FLAG
This instruction tests for "zero" in (Y). If all bits are zero, they are set to "ones" and the next instruction is skipped. If any bit is not "zero," the next instruction is executed and (Y) is set to "ones". This instruction, in conjunction with programming conventions, provides control over simultaneous processing of shared memory data. This information utilizes the Type II General Format.

77 50 ENTER ABSOLUTE PAGE REGISTER
This instruction places the least significant bits of Y into the Worker Absolute Page Register designated by $\hat{j}$ of the instruction word. This instruction utilizes the Type II Special Format.

77 54 STORE ABSOLUTE PAGE REGISTER
This instruction stores the bits of the Worker Absolute Page Register specified by $\hat{j}$, at the Y address. This instruction utilizes the Type II Special Format.

77 63 LOAD B7 AND JUMP
This instruction stores P in the B^7 register and jumps to address (Y). This allows procedure calls to programs in shared memory without the danger of losing the return address. This instruction utilizes the Type II General Format.

77 64 LOAD B^7, JUMP AND SET DIRECT MODE
This instruction stores P in the B^7 register and jumps to address (Y). This allows procedure calls to programs in shared memory without the danger of losing the return address. This instruction utilizes the Type II General Format. Before executing the jump, set the addressing mode to Direct addressing.

77 65 LOAD B^7, JUMP AND SET PAGE MODE
This instruction stores P in the B^7 register and jumps to address (Y). This allows procedure calls to programs in shared memory without the danger of losing the return address. This instruction utilizes the Type II General Format. Before executing the jump, set the addressing mode to Page addressing.

77 00 Fault

77 77 Fault

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APPENDIX B
THIN-FILM INPUT AMPLIFIER
(FOR REFERENCE ONLY)

10. SCOPE

This document covers design requirements for thin-film digital input amplifiers as described herein.

20. APPLICATION

The circuit should provide the equivalent performance of the present discrete input amplifier as shown in figure 1. This circuit will be used at the termination of 0-100 feet of twisted pair cable, which is one of four cables being driven by one data line driver as shown in figure 2.

30. CIRCUIT INPUT CHARACTERISTICS

30.1 Voltage Level Inputs - The binary "one" state shall be 0 volts $\pm$ 0.5 volts and the binary zero shall be 4 volts $\pm$ 1 volt as measured at the output terminal of the data line driver (see figure 2).

30.2 Trigger Threshold Level - The threshold level distinguishing the "one" state from the "zero" state is + 1.5 volts $\pm$ 0.7 volts at the input amplifier terminal.

30.3 Noise Rejection - The circuit should be capable of $\pm$ 5 volts common mode noise rejection. It should be capable of handling noise levels up to $\pm$ 5 volts peak, without any change in output level.

30.4 Rise and Fall Times - The time required for a data line driver to switch logic levels, as measured from the 10 percent to 90 percent of full amplitude points at the output terminals of the driver (before the 100 feet of twisted pair cable), shall not be greater than 1.0 microsecond.

30.5 Loading - The maximum steady state current drawn from a line by an input amplifier circuit shall not exceed 2.5 milli-amps at 4.0 volts.

30.6 Input Impedance - The input amplifier circuit must have an ac input impedance of 130 ohms and dc input impedance of 2000 ohms.

40. TEMPERATURE

The circuits must be capable of meeting all the requirements of this specification over the full military temperature range of -55°C to +125°C.

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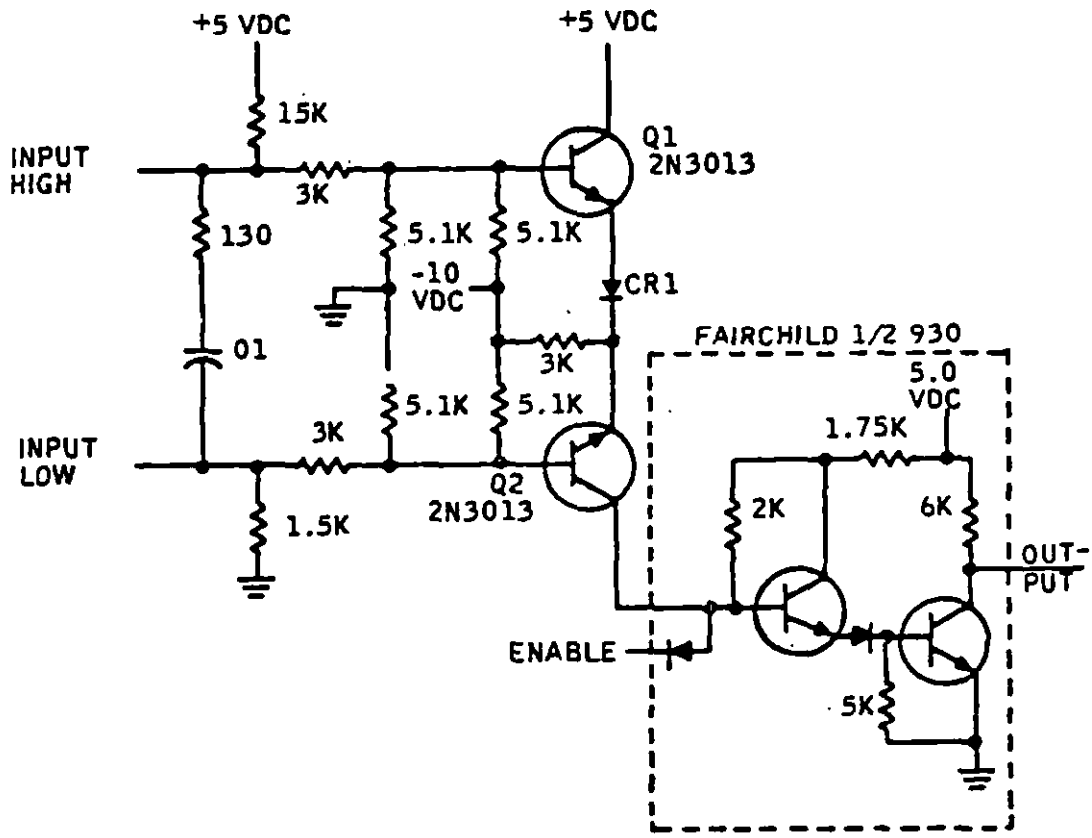


Figure 1. Typical Input Amplifier Circuit
(FOR REFERENCE ONLY)

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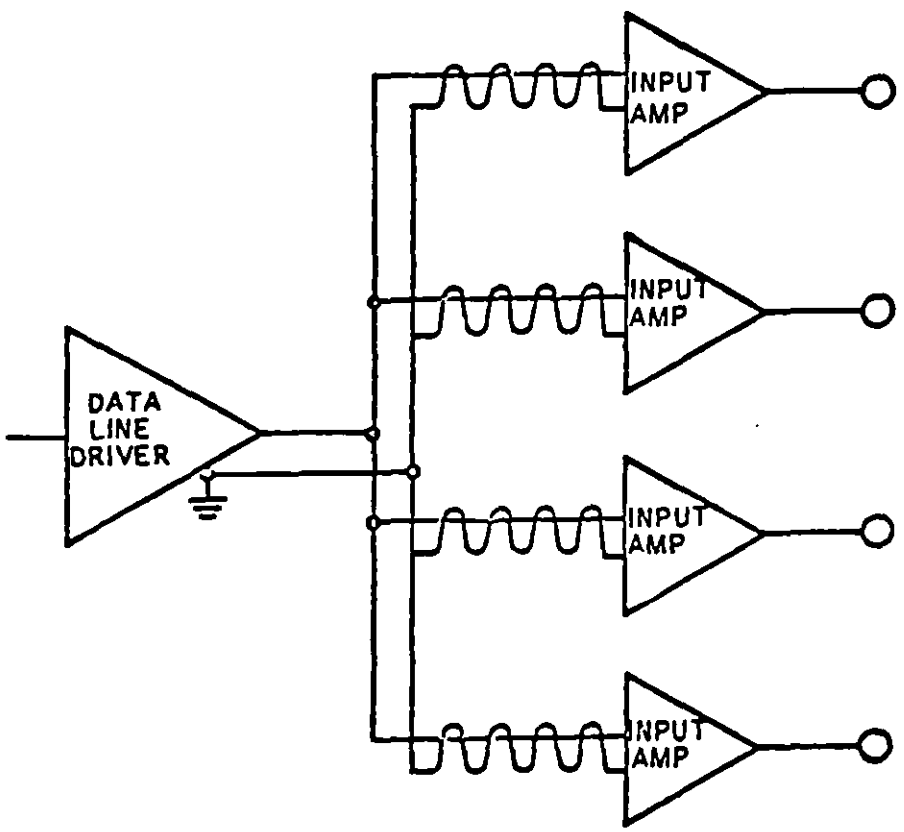


Figure 2. Input/Output Drivers

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